

Unit 10 Overview: Representing Data and Distributions

Unit Narrative



In this unit, students will extend their learning from Unit 9 to explore the impact of outliers on data distributions. Students examine features of data sets to determine appropriate measures of center and spread and eventually compare and create their own graphical representations of real-world data. Using equivalent ratios, students will also learn how to make predictions about populations from sample data.

Section 1 focuses on the impact data values have on the measures of center and variation. Students will use features of data sets to justify which measures best summarize the distribution. When determining the appropriate measure of center, students will use mean and median. When determining the appropriate measure of variation, students will use the range or interquartile range to describe the data distribution. Visuals are critical for this section to make sure students can see the shape of the distribution when determining the most appropriate measures of center and variation to describe the data. In Section 2, students compare data sets and graphs for different sets of data.

Section 3 introduces students to what a scale factor is and how it can be used in making predictions about populations from sample data. Scale factor is purposely explained as a ratio because students will work at identifying equivalent ratios throughout these lessons, providing an opportunity to make explicit connections to ratio reasoning from Unit 5. Once they've identified equivalent ratios and scale factors, students will work to understand how a sample can be used to make predictions about a larger population. Students are exposed to real-world situations to truly understand scale factors in this situation. The strategies highlighted within the lessons in this section do not include setting up or solving proportions. Proportions will be addressed and formalized in Unit 12. Teachers are encouraged to model strategies like those shown in this unit as students build a stronger conceptual understanding of ratios and proportional relationships prior to the next course.

Unit At-A-Glance

Section 1: Data Impact on Measures of Center and Variation	Benchmarks Addressed	Lesson 1	Matching Data Displays
	MA.6.DP.1.5 MA.6.DP.1.6 MA.7.DP.1.1	Lesson 2	Outliers
		Lesson 3	Justifying Measures of Center and Variation
		Lesson 4	Determining Appropriate Measures of Center and Variation
Section 2: Comparing Data Sets	Benchmarks Addressed	Lesson 5	Comparing Graphical Representations
	MA.7.DP.1.2	Lesson 6	Comparing Data Sets
		Lesson 7	Comparing Data Sets and Graphs
Section 3: Scaling Sample Data to Populations	Benchmarks Addressed	Lesson 8	Random Samples vs. Population
	MA.7.DP.1.3	Lesson 9	Making Population Predictions – Part 1
		Lesson 10	Making Population Predictions – Part 2



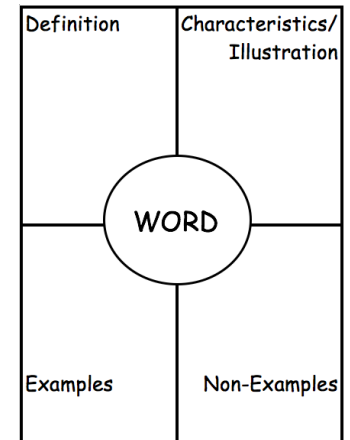
Differentiation

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Engagement <i>The “WHY” of learning</i>	• Allow students to input data into an Excel spreadsheet to create graphs and charts. Students can use the equation features to manipulate data to find the different features of a data set. • Multiple Wrap-Up components in this unit provide students with the opportunity to take ownership of their learning as they self-assess their understanding of the learning targets.
Representation <i>The “WHAT” of learning</i>	• Students use visual models throughout the unit to model and compare data sets. Auditory entry points are available through partner and small-group discussion throughout each lesson. • Provide auditory cues when introducing and using key words.
Action & Expression <i>The “HOW” of learning</i>	• Provide students with access to assistive technologies, such as audio recorders, speech-to-text software, and virtual manipulatives. This will be particularly helpful during some of the Wrap-Up components in which students must explain their responses, to provide an additional entry point and means of assessing understanding. • As students are navigating the lessons in Section 3, highlight the different ways that students can use equivalent ratios to solve problems to make connections to prior content.

ELL Information

Students will be identifying and interpreting key features of each data representation and choosing graphs to represent numerical and categorical data. This unit provides an opportunity to review some important vocabulary. Have students create a Frayer Models for the words in this topic:



- A Frayer Model (similar to the four-square graphic organizer) is a helpful graphic organizer because it allows students to see terms in multiple ways. In this unit, students review graphical representations of data and determine which graphs best represent different data sets.
- Provide the definition, but have students work in pairs to illustrate the word (when possible, or else list characteristics), list examples of the types of data sets that are best represented, and list non-examples.
- For examples and non-examples, encourage students to consider this a work in progress, and to add throughout the section as they are presented.
- Consider creating a Frayer Model for each term from the Glossary by Section lists (below).

All glossary terms are available in multiple languages, including English, Spanish, Haitian Creole, and American Sign Language, as support for ELL students. Consider pairing the following videos with strategies from “Additional Differentiated Support.”

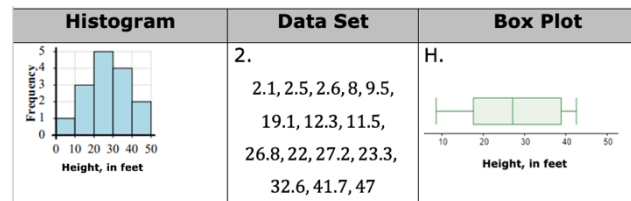
- | | | |
|--|---|--|
| <ul style="list-style-type: none"> • Box plot • Histogram • Mean • Measures of variability • Population (in data analysis) • Range (of a data set) | <ul style="list-style-type: none"> • Data • Interquartile range (IQR) • Median • Mode • Quartile • Stem-and-leaf plot | <ul style="list-style-type: none"> • Frequency table • Line plot • Measures of center • Outlier • Random sampling |
|--|---|--|

Additional Differentiated Support

Scaffolding Resources

Card Sort

Lesson 1 engages students in a card sort activity to make sense of data through a matching activity. Students work collaboratively to match data presented to the corresponding histogram and box plot. Through this card sort, students can draw connections between data presented in tables, box plots, and histograms to deepen understanding using multiple representations.



Partner Reading Strategy

- Prior to the lesson, teacher will find appropriate text-heavy word problems that address using expressions to represent real-world situations.
- Working in pairs, one student reads the word problem aloud, while the other student assists in pronunciation or assistance when needed.
- When both students have practiced reading and received feedback, they will begin solving the problem together to find the solution.

Whenever possible, act out or provide illustrations of word problems to help students visualize the problem they are going to solve.

- Read problems out loud and gesture or point to the illustration to reinforce the meaning.

For some problems, students can draw their own pictures. Try having them do this after you have shown them some examples.

On-Ramp

On-Ramp to 6th Grade Math provides video and practice tools for students to scaffold the following skills:

- Creating and Interpreting Line Plots
- Creating Line Plots with Fractions
- Using Line Plots to Solve Problems



Section Overview

Section 1: Data Impact on Measures of Center and Variation (Lessons 1-4)

Benchmarks Addressed	Learning Targets	
MA.6.DP.1.5 Create box plots and histograms to represent sets of numerical data within real-world contexts. <i>Clarification 1: Instruction includes collecting data and discussing ways to collect truthful data to construct graphical representations.</i> <i>Clarification 2: Within this benchmark, it is the expectation to use appropriate titles, labels, scales and units when constructing graphical representations.</i> <i>Clarification 3: Numerical data is limited to positive rational numbers.</i> MA.6.DP.1.6 Given a real-world scenario, determine and describe how changes in data values impact measures of center and variation. <i>Clarification 1: Instruction includes choosing the measure of center or measure of variation depending on the scenario.</i> <i>Clarification 2: The measures of center are limited to mean and median. The measures of variation are limited to range and interquartile range.</i> <i>Clarification 3: Numerical data is limited to positive rational numbers.</i> MA.7.DP.1.1 Determine an appropriate measure of center or measure of variation to summarize numerical data, represented numerically or graphically, taking into consideration the context and any outliers. <i>Clarification 1: Instruction includes recognizing whether a measure of center or measure of variation is appropriate and can be justified based on the given context or the statistical purpose.</i> <i>Clarification 2: Graphical representations are limited to histograms, line plots, box plots and stem-and-leaf plots.</i> <i>Clarification 3: The measure of center is limited to mean and median. The measure of variation is limited to range and interquartile range.</i>	Lesson 1	I can match displays of data to the appropriate data set.
	Lesson 2	- I can determine the implications an outlier has in a given set of data. - I can relate the impact of an outlier to which measures of center and variation best describe a distribution.
	Lesson 3	I can justify whether or not a measure of center or variation appropriately summarizes data.
	Lesson 4	I can determine an appropriate measure of center and variation to summarize numerical data.

Section 2: Comparing Data Sets (Lessons 6-7)

Benchmarks Addressed	Learning Targets	
MA.7.DP.1.2 Given two numerical or graphical representations of data, use the measure(s) of center and measure(s) of variability to make comparisons, interpret results and draw conclusions about the two populations. <i>Clarification 1: Graphical representations are limited to histograms, line plots, box plots and stem-and-leaf plots.</i> <i>Clarification 2: The measure of center is limited to mean and median. The measure of variation is limited to range and interquartile range.</i>	Lesson 5	I can use two graphical representations of data to make comparisons and draw conclusions about the populations.
	Lesson 6	I can compare and interpret two numerical representations of data for two populations.
	Lesson 7	I can compare and interpret a numerical representation of data of one population and a graphical representation of another population.

Section 3: Scaling Sample Data to Populations (Lessons 8-10)

Benchmarks Addressed	Learning Targets	
MA.7.DP.1.3 Given categorical data from a random sample, use proportional relationships to make predictions about a population.	Lesson 8	I can determine if a sample is representative of the population.
	Lesson 9	I can use proportional relationships to make predictions about a population.
	Lesson 10	I can use proportional relationships to make predictions about a population.



Vertical Alignment

Grade 6 Accelerated Unit 10
Representing Data and Distribution



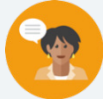
Grade 7 Accelerated Unit 13
Representing Data

Prior Course(s)	Course Level	Future Course(s)
MA.5.DP.1.1 Collect and represent numerical data, including fractional and decimal values, using tables, line graphs, or line plots. MA.5.DP.1.2 Interpret numerical data with whole-number values, represented with tables or line plots, by determining the mean, mode, median, or range.	MA.6.DP.1.5 Create box plots and histograms to represent sets of numerical data within real-world contexts. MA.6.DP.1.6 Given a real-world scenario, determine and describe how changes in data values impact measures of center and variation. MA.7.DP.1.1 Determine an appropriate measure of center or measure of variation to summarize numerical data, represented numerically or graphically, taking into consideration the context and any outliers. MA.7.DP.1.2 Given two numerical or graphical representations of data, use the measure(s) of center and measure(s) of variability to make comparisons, interpret results, and draw conclusions about the two populations. MA.7.DP.1.3 Given categorical data from a random sample, use proportional relationships to make predictions about a population.	MA.7.DP.1.5 Given a real-world numerical or categorical data set, choose and create an appropriate graphical representation. MA.912.DP.1.4 Estimate a population total, mean or percentage using data from a sample survey; develop a margin of error through the use of simulation.

10.1 Matching Data Displays

Lesson Introduction

 Benchmark	MA.6.DP.1.5 Create box plots and histograms to represent sets of numerical data within real-world contexts.		 Learning Target	I can match displays of data to the appropriate data set.
 Benchmark Coherence	Building On		Working Toward	
	MA.5.DP.1.1 Collect and represent numerical data, including fractional and decimal values, using tables, line graphs, or line plots.		MA.7.DP.1.5 Given a real-world numerical or categorical data set, choose and create an appropriate graphical representation.	
 Essential Questions	- What is the difference between a box plot and histogram? - How can you correctly match a data set to a box plot? - How can you correctly match a data set to a histogram? - Which types of data are better to use with box plots and histograms?			
 Lesson Narrative	In this lesson, students begin by finding the five-number summary of a data set, then choose the box plot and histogram that displays the data correctly. Students explain which features of each display they utilized to find the match and then compare the box plot and histogram to decide which displays the data better. Students then match histograms to data sets and box plots before wrapping up with a growth mindset reflection.			
 Component Summary	10.1.1 Warm-Up (~5 min.)	Find the five-number summary and range for a set of data (<i>SE p. 105</i>)	10.1.3 Activity (15 min.)	Sort cards to match data set with histogram and box plot (<i>SE p. 107</i>)
	10.1.2 Exploration (15 min.)	Find the five-number summary and IQR with a partner to create box plot and histogram (<i>SE p. 106</i>)	10.1.4 Your Turn! (5 min.)	Find the five-number summary with IQR and match the data points to the correct box plot and histogram (<i>SE p. 108</i>)
	10.1.5 Wrap-Up (~5 min.)	Students reflect on their growth mindset throughout the lesson (<i>SE p. 108</i>)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Box plot - Histogram		- Card sort (Optional) Additional paper (Optional) Computer	- The card sort will need to either be printed and cut or digitally altered to be given to each student. If this is not possible, project the cards in a way for the whole class to see them. - Consider creating partner-pairs before the lesson begins. - Before the lesson is presented to students, decide how and in what format the card sort will be presented.

 Teacher Tips	Common Misconceptions - Students may incorrectly match the data to a display by misreading the data. - Students may calculate or interpret the IQR incorrectly. - Students may not understand that certain information cannot be determined from a box plot/histogram.	Things to Consider - Consider encouraging students to create their own box plots and histograms based on data sets before choosing their answers if they get stuck. - Consider reviewing the features of the five-number summary to activate prior knowledge before students begin the lesson.
	Literacy and Language Routines Notice and Wonder 10.1.3 Activity provides students with the opportunity to make connections between data sets, histograms, and box plots. Engage students in a Notice and Wonder routine before they begin matching to encourage students to draw conclusions about the data and consider what the shape, center, and spread should look like. This will help students accurately match the visual displays and data sets.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.2.1: Multiple Representations Students analyze and present data in a variety of ways throughout this lesson. Clarifications for this standard includes showing students that various representations can have different purposes and can be useful in different situations. This is particularly evident as students consider what information can be drawn from box plots versus histograms, understanding that both representations may be better suited for different purposes.
	 Differentiate Instruction	
Lesson Notes:		

10.1.1 Warm-Up: Bell Work

Component Overview: Students determine the five-number summary and range for a set of data in context.



10.1.1 Warm-Up: Bell Work

1. During the 2020 NBA Season, Jimmy Butler played in the Championship Game against the Los Angeles Lakers. The total points scored by Jimmy Butler for the conference semifinals, conference finals, and championship games are listed below.

12, 35, 22, 40, 25, 23, 22, 17, 24, 17, 14, 20, 17, 17, 30, 13, 40

- a. Determine the five-number summary for the data set.

Min: 12

Q1: 17

Median: 22

Q3: 27.5

Max: 40

- b. Determine the range.

28



Observe students as they work to determine if additional prompting or support may be needed. Some students may forget that it is helpful to first rewrite the data set in order from least to greatest when determining the five-number summary.

Consider asking the following questions, if needed:

- What are the five measures included in the five-number summary?
- When determining the five-number summary and range, what is helpful to do with the values in the data set first?

10.1.2 Exploration: Matching Displays

Component Overview: In this Exploration, students will work with a partner to find the five-number summary and the interquartile range for the given set of numbers. With their partner, students will determine which displays correctly represent the data set and discuss which display is best to represent the given data.



10.1.2 Exploration: Matching Displays

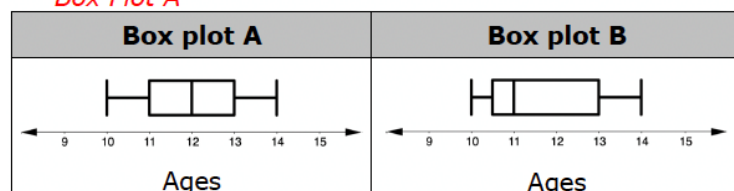
Work with your partner to complete the following:

A middle school band director collects the ages of each clarinet player in the band, shown below.

10, 10, 11, 11, 11, 12, 12, 12, 13, 13, 14

- Determine the five-number summary and IQR for the ages of the clarinet players in the band.
*Minimum: 10, Maximum: 14, Quartile 1: 11, Median: 12
Quartile 3: 13, IQR: 2*

- Which box plot accurately displays the ages of the clarinet players?
Box Plot A



- Discuss your choice with your partner and summarize.
Answers will vary. The correct box plot can be quickly identified because the median is incorrect in Box plot B.

In a box plot, whiskers extend from each quartile to the minimum or maximum. The left whisker represents the bottom 25% of the data and the right whisker represents the top 25%.

- Discuss with your partner how the location of the median and the length of the whiskers can be used to determine the shape of the distribution. Summarize your discussion.
Answers will vary. If the length of each whisker is about the same, the shape is roughly symmetric. If one whisker is longer than the other, the shape is skewed.



MA.K12.MTR.2.1: Multiple Representations

In this activity, students will work with a partner to determine appropriate representations of the given data in different ways, including: a box plot, a table, and a histogram.



For students who are struggling to explain how the median and whiskers can be used to determine the shape of the distribution, ask questions like:

- What does the median tell you about the data set? How does this impact the shape of the distribution?
- What do the length of the whiskers tell you about the data set? How does this impact the shape of the distribution?

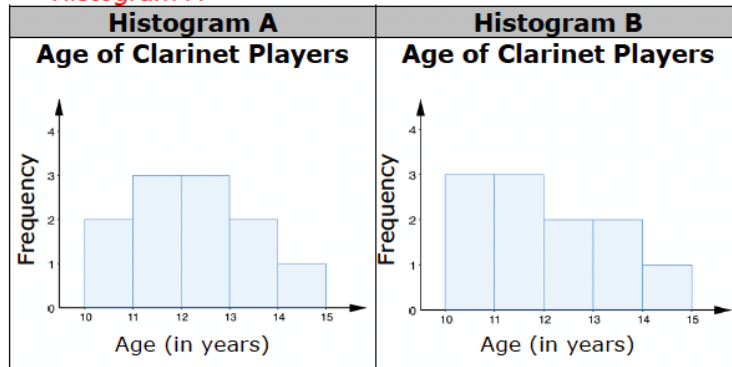
10.1.2 Exploration: Matching Displays (continued)

5. Create a frequency table of the data set.

Age Range	Number of Clarinet Players
10 – 11	2
11 – 12	3
12 – 13	3
13 – 14	2
14 – 15	1

6. Which histogram most accurately displays the data?

Histogram A



7. Discuss which characteristics of the histogram you used to make your selection.

Frequency data.

8. Describe the shape of the histogram you chose, including any gaps, outliers, or clusters.

Answers will vary. Histogram A has a bell-shaped curve. The data is evenly distributed. It is clustered about ages 11 – 13 with no real outliers.

9. Compare the box plot selected for the data with the histogram selected. Discuss which display better represents the data. Summarize your discussion.

Answers will vary. The box plot represents the data better because you can see the ages are clustered around 11 – 13. The histogram represents the data better because you can see the frequency of the ages of student clarinet players.



This activity provides multiple entry points through visual displays of data and partner discussion.



Challenge students to come up with examples of types of data that would be better displayed in a box plot or histogram and explain why.

10.1.3 Activity: Card Sort

Component Overview: In this activity, students work collaboratively to match histograms with the raw data sets used to create them and box plots that represent the same data through a card sort. If printing the cards is not an option, project the cards for the whole class to see.



10.1.3 Activity: Card Sort

1. An arborist visited a pine tree farm and took measurements of the tree heights, in feet, in three different fields. He dropped his paperwork and all of the data he collected got mixed up.

Complete the table to match each histogram with its corresponding data set and box plot. Write the number and letter from the card that corresponds to each match.

Histogram	Data Set	Box Plot
Frequency Height (in feet)	1	H
Frequency Height (in feet)	2	B
Frequency Height (in feet)	3	Y

2. Compare your answers with your partner. Make any revisions, if needed.



Notice and Wonder

Before the card sort, consider engaging students in a Notice and Wonder routine. Encourage students to draw conclusions about the data by asking questions like:

- What features did you use to match the displays and data?
- Did you start by matching data to a display or matching the displays to each other? Why?



The card sort will help provide kinesthetic learners with an entry point into the lesson. If students are matching incorrectly, consider prompting to justify their choices.

10.1.4 Your Turn!

Component Overview: In 10.1.4 Your Turn!, students work independently to match a box plot and histogram to a given set of data.

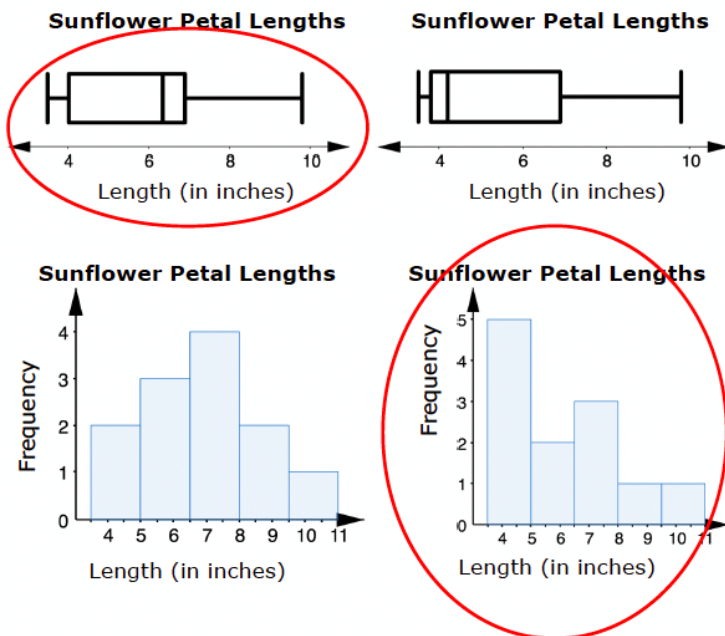


10.1.4 Your Turn!

1. A horticulturist is researching sunflowers to find out the lengths of the petals on a sunflower. She gathers information from 12 sunflowers and measures the length of the petals, in inches, on each flower.

6.8, 6.5, 6.2, 3.5, 3.6, 4.1, 3.8, 4.5, 6.7, 7.2, 8.1, 9.8

Circle the boxplot and the histogram that best display the data set.



To ensure students are thinking through each choice and not simply guessing, ask questions like:

- What is the median of the data set? How do you know? (Finding the median will allow students to identify the correct box plot.)
- How would you describe the spread of this data? Is it symmetrical? Why or why not? (When writing the data in order, students may observe that there are many small values in comparison to larger values to help identify the correct histogram.)



For students who finish early, ask them to explain which display better represents the data and why. This could also include asking which features of the data are made most clear in each representation.

10.1.5 Wrap-Up: Growth Mindset

Component Overview: Students will reflect on their growth mindset throughout the lesson. Encourage students to answer with fidelity to find areas of weakness that can be targeted in a later lesson.



10.1.5 Wrap-Up: Growth Mindset

1. Complete the table as you reflect on today's lesson.

Answers will vary.

Which mindset did I demonstrate?	Check the best description.
Did I use the partner discussions as learning opportunities?	☐ Never ☐ Sometimes ☐ Usually ☐ Always
Did I work on problems that challenged me?	☐ Never ☐ Sometimes ☐ Usually ☐ Always
Did I use strategies to "un-stick" myself when I found the problems difficult?	☐ Never ☐ Sometimes ☐ Usually ☐ Always



To provide additional means of observation and reflection, encourage students to find and name examples for the mindsets they demonstrated. For example, pointing out a strategy they used in the lesson to "unstick" themselves.

10.2 Outliers

Lesson Introduction

 Benchmarks	MA.6.DP.1.6 Given a real-world scenario, determine and describe how changes in data values impact measures of center and variation. MA.7.DP.1.1 Determine an appropriate measure of center or measure of variation to summarize numerical data, represented numerically or graphically, taking into consideration the context and any outliers.		 Learning Targets	- I can determine the implications an outlier has in a given data set. - I can relate the impact of an outlier to which measures of center and variation best describe a distribution.
 Benchmark Coherence	Building On MA.5.DP.1.2 Interpret numerical data with whole-number values, represented with tables or line plots, by determining the mean, mode, median, or range.		Working Toward MA.912.DP.1.1 Given a set of data, select an appropriate method to represent the data, depending on whether it is numerical or categorical data and on whether it is univariate or bivariate.	
 Essential Questions	- What is an outlier? - How does an outlier affect measures of center and variation? - Are there times when an outlier does not affect the measures? - What measures of center and variation can be used to describe sets of numerical data? - How does removing an outlier from a data set affect the measures of center and variation?			
 Lesson Narrative	This lesson begins with students exploring data presented in a line plot. Students analyze the information presented and apply what they know about measures of center and variation to describe and summarize the data, considering the impact an outlier has on the data set and what happens once it is removed. Students summarize the impact of outliers on measures of center and variation, applying this understanding to synthesize data presented in a histogram and other data displays.			
 Component Summary	10.2.1 Warm-Up (~5 min.)	Watch a video and reflect on the impact of errors (<i>SE p. 109</i>)	10.2.4 Guided Practice (10 min.)	Calculate impact of outlier on data set (<i>SE p. 112</i>)
	10.2.2 Exploration (10 min.)	Work with a partner to find the five-number summary and IQR (<i>SE p. 110</i>)	10.2.5 Exploration (10-15 min.)	Discover how to find best fit measurement of variation and center (<i>SE p. 113</i>)
	10.2.3 Bring It Together (5 min.)	Synthesize how outliers affect data (<i>SE p. 111</i>)	10.2.6 Your Turn! (5 min.)	Analyze how an outlier impacts a data set (<i>SE p. 115</i>)
	10.2.7 Wrap-Up (~5 min.)	Reflect on understanding of outlier impact on data (<i>SE p. 115</i>)		
 Resources	Glossary		Materials	
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Outlier - Interquartile range		(Optional) Paper (Optional) Tape and labels for dot plot (Optional) Computer	
		Additional Preparation 10.2.1 Warm-Up contains a video to accompany the question prompts.		

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may incorrectly determine an outlier can only be higher or only be lower than the rest of the data. - Students may think that an outlier will always significantly change the mean. - Students may struggle to draw conclusions about the effect of outliers on measures of center and variation. - Students may think that removing outliers will always have the same impact on the data set, regardless of whether the outlier falls above or below the norm in the set.	- Consider providing examples and nonexamples of data sets containing outliers to students to decipher the differences. - Allowing students to use calculators during the lesson will help students focus on the effect of outliers on measures of center and variation. Consider have no-calculator questions where students turn their calculator over. - If time is an issue, then consider assigning 10.2.6 Your Turn! for homework.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Think-Pair-Share 10.2.4 Guided Practice provides an opportunity for students to engage in a Think-Pair-Share. This routine encourages students to synthesize information as they first consider the question and then discuss with a partner to build onto or revise their thinking. Through this routine, students have the chance to deepen understanding of the impact of outliers.	MA.K12.MTR.7.1: Real-World Contexts Students will be making connections between data displayed on graphs and histograms and real-world contexts throughout this lesson. Students will use the information presented to determine measures of center and variation and connect this back to the context of the problems to reason about the impact of outliers when summarizing data distributions.
 Differentiate Instruction	Utilizing various display methods to show the data provides a variety of visual entry points for learners. Students can engage with the content through histograms, dot plots, and tables. Encourage students to create their own data displays to help answer questions about the data if you notice them struggling to conceptualize the data presented without a visual display. Students work collaboratively throughout the exploration and guided practice, providing opportunity for mathematical discourse as well as auditory entry points.	

Lesson Notes:

10.2.1 Warm-Up: Precision Is Key

Component Overview: Play the Career Profile video on the quality control engineer for the entire class. After students watch the video, prompt them to answer the two questions independently. Consider stopping the video at minute 2:22 to allow students enough time to reflect on and complete the questions.



10.2.1 Warm-Up: Precision Is Key

Becky Miller is a quality control engineer with GE Aviation. Like many other careers, manufacturing engineering has many moving parts that make up an entire process. Becky says that manufacturing is problem-solving.

1. List at least two steps that you take when faced with a difficult task.

Answers will vary. Understand the problem. Make a plan. Keep trying.

Becky mentions that an error “about the size of a human hair” can make a difference in manufacturing jet engines.

2. What is a time that you have made a small error that caused a big impact in your next steps?

Answers will vary. I used the wrong inverse operation when solving an equation. I woke up late for school.



This Warm-Up positions students to consider the importance of precision when working with numbers in mathematics. To connect to the rest of the lesson, ask questions like:

- *Would rounding numbers to the wrong decimal place ever impact the correctness of your answer?*
- *When thinking about answers would you ever say that 141.3 phone calls were made?*
- *When speaking of money is it accurate to write \$5.6?*

10.2.2 Exploration: Determining, Describing, and Comparing Numerical Data Represented Graphically

Component Overview: Students will work with a partner to calculate the five-number summary and IQR of a data set. Students then remove the outliers from the set and recalculate to find the effects an outlier has on the values used to summarize the distribution. Encourage students to make connections between measures that change and do not change when the outlier is removed.

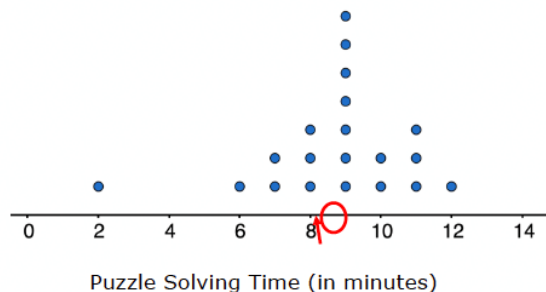


10.2.2 Exploration: Determining, Describing, and Comparing Numerical Data Represented Graphically

Work with your partner to complete the following.

- Twenty students timed how long it took each of them to solve a puzzle. Below is a list of their solution times, in minutes, and a dot plot displaying the data.

2, 6, 7, 7, 8, 8, 8, 9, 9, 9, 9, 9, 9, 9, 10, 10, 11, 11, 11, 12



- Describe the distribution of data.
Answers will vary. The data is symmetrical if not for the outlier with the value of 2. The outlier causes the data to skew to the left.
- Determine the mean of the data set. Then, draw an arrow (↑) below the number line that points to this value.
8.7
- Determine the median of the data set. Then, circle the value on the number line that represents this value.
9
- Discuss with your partner what you notice about the mean and the median of the data. Summarize your discussion.
Answers will vary. The mean and median are very close to each other. The mean is slightly less than the median.



While students are exploring in small groups or pairs, monitor for the use of both informal and formal language in their descriptions. Use their responses as probing questions for 10.2.3 Bring It Together for meaningful synthesis.

For example, you may notice Student A describing the distribution as “normal-looking” or using similar informal language, while Student B in another group may describe it as symmetrical.

As you take note of student responses here, see possible question stems to include in 10.2.3 Bring It Together.



Have students make a connection between the outlier and the median and mean.

10.2.2 Exploration: Determining, Describing, and Comparing Numerical Data Represented Graphically (continued)

e. What is the range of the data?

10

f. Find the interquartile range (IQR).

Q1	Q3	IQR
8	10	2

2. Imagine that the fastest time of 2 minutes was removed from the data set. Discuss with your partner what effect, if any, removing this value has on the values of center and variation (mean, median, range, IQR). Summarize your discussion.

Answers will vary. Removing the fastest time will make the graph a more accurate description of the time it takes to complete the puzzle.

3. How does the shape of the graph change after 2 is removed?

Answers will vary. The graph would be more symmetrical. The graph will be more representative of the time it takes to complete the puzzle.

4. Compare the new values with the original values.

	Original Value	>, <, =	New Value after 2 is Removed
Mean	8.7	<	9.1
Median	9	=	9
Range	10	>	6
IQR	2	=	2

5. Which measure(s) of center will be affected if the outlier is removed?

- ☐ mean
- ☐ median
- ☐ both the mean and median

6. Which measure(s) of variation will be affected if the outlier is removed?

- ☐ range
- ☐ interquartile range
- ☐ both the range and the interquartile range



Notice and Wonder

Implement a Notice and Wonder routine to encourage students to notice what happens to each measure of center and variation after the outliers have been removed.

- What do you notice about the original values compared to the new value of each category after 2 is removed?
- What do you notice about the two categories that stayed the same?
- What do you wonder? Do you think this happens when outliers are removed from other sets of data?

10.2.3 Bring It Together: Outlier Effects

Component Overview: Students complete statements to describe how the presence of outliers impact the measures of center and variation.



10.2.3 Bring It Together: Outlier Effects

1. Complete each statement to describe the effect the outlier, 2, has on each measure of center or variation.

- The presence of an outlier does not have a significant effect on the ☐ mean.
☒ median.
- The presence of an outlier will change, often significantly, the value of the ☒ mean.
☐ median.
- The presence of an outlier does not have a significant effect on the ☒ IQR.
☐ range.
- The presence of an outlier will change, often significantly, the value of the ☐ IQR.
☒ range.



When outliers are present or a distribution is skewed, the **median** is the preferred measure of center and the **IQR** is the preferred measure of variation to describe the distribution.



Use inquiry and talk moves to have students make sense of their learning with the sentence frames and the blue box, using questions to have them formalize their descriptions from the Exploration.

- Student A, you described the data distribution as 'normal-looking' – can you say more about what you mean by that?
- Student B, how did you describe the distribution?
- Student A, based on Student B's response, would you change your answer to the sentence frame? Why or why not?
- Student B, what impact does the outlier have on the distribution?
- Student A, based on that response, should we use median or IQR in the Exploration? Why?

10.2.4 Guided Practice: Data Outliers

Component Overview: In this component, students work with a partner to apply their knowledge of calculating the five-number summary and IQR to determine how an outlier can impact the data. Students remove an outlier to determine how it impacts a data set and the measures of center and variation.



10.2.4 Guided Practice: Data Outliers

1. Zoe needed exact change to purchase a snack and a drink from the vending machine. She asked her friends how much change, in dollars, they had and listed the results.

0.15, 0.15, 0.20, 0.25, 0.25, 0.30, 0.30, 0.30, 0.35, 0.75, 0.85

- a. What do you notice about the data?
Answers will vary. All numbers are less than \$1.
- b. Will the mean be higher, lower, or approximately the same as the median? Explain your reasoning or justify your answer by showing your work.
Answers may vary. Slightly higher. Mean is 0.35 and the median is 0.30.
- c. Describe the impact of removing 0.85 from the data set.
Answers may vary. The data becomes more symmetrical. Now 0.75 is the only outlier. The mean will decrease.
- d. The values 0.85 and 0.75 are outliers in this data set. What happens to the range if 0.85 and 0.75 were both removed from the data set?
The range becomes smaller and the spread of the data is smaller
- e. The IQR of the full data set is 0.15.

The IQR with 0.85 and 0.75 removed is 0.125.
- f. Is there a big difference between the IQR of the original data set and the modified data set?
No, the IQR is not significantly affected by the removal of outliers



Consider providing a graphic organizer or table for students to use when finding the mean, median, and interquartile range.



To incorporate digital learning, allow students to type the numbers into a column on an Excel spreadsheet to manipulate the data and view how changing the different results changes the data.



Think-Pair-Share
Engage students in a Think-Pair-Share routine to complete this problem. This will provide students with the opportunity to formalize their thinking and synthesize the impact of outliers in a data set and on measures of center and variation.

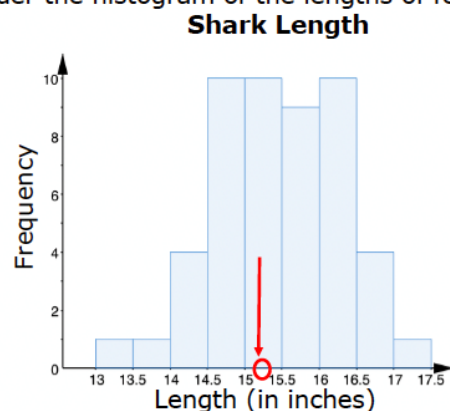
10.2.5 Exploration: Measures of Center

Component Overview: In this exploration, students work with a partner to determine the measures of center and spread that best describe the data presented in two different histograms. The first histogram is relatively symmetrical, while the second is skewed to the right. Students analyze the shape of the histograms to draw conclusions about how outliers impact the data sets. Students will also evaluate the best measures of center and spread to use when describing symmetrical versus skewed data sets.



10.2.5 Exploration: Measures of Center

1. Consider the histogram of the lengths of female sharks.



- a. Describe the shape of the histogram.
The graph is symmetrical, with most sharks having a length between 14.5 and 16.5 inches.
- b. Circle the value on the horizontal axis of the graph that shows the approximate location of the median of the data.
Shown on graph.
- c. Draw an arrow ($\uparrow$) on the graph to indicate the approximate location of the mean of the data.
Shown on graph.
- d. Compare your answers to B and C with your partner's answers. Discuss your reasoning, and if necessary, make changes.



The histogram provides a visual entry point. Consider having students further distinguish between median and mean by using different colored pencils to label each on the graph.



In a symmetrical graph, the mean and the median are located in the middle of the graph, approximately in the same location.

10.2.5 Exploration: Measures of Center (continued)

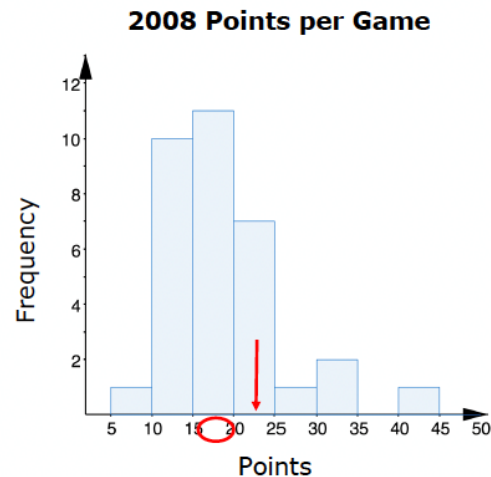
- e. The mean and the median of this histogram are located between 15 – 15.5. Discuss with your partner which measure most accurately describes the center of the histogram. Summarize your discussion.

Answers will vary. Both measures have about the same value, but the mean best describes the data because the average represents a typical length.



In a symmetrical graph with no outliers, the **mean** is the preferred measure of center and the **range** is the preferred measure of variation to describe the distribution.

2. The histogram displays the number of points per game scored by a college basketball player during the 2008 season.



- a. Describe the shape of the histogram.
Skewed to the right.



To solidify conceptual understanding, read all three blue boxes in 10.2.5 as a class and ask the following questions:



- Are the values for mean and median usually close together?
- What could cause them to be farther apart?
- Why is the mean the preferred measure of center when a data set has no outliers?
- Why is the range the preferred measure of variation to use with mean?



Consider allowing students to draw pictures to match the words in the second blue box for those who would benefit from a visual of the statement.

10.2.5 Exploration: Measures of Center (continued)

- b. Circle the value on the horizontal axis of the graph that shows the approximate location of the median of this data.

Shown on graph.

- c. Draw an arrow ($\uparrow$) on the graph to indicate the approximate location of the mean of the data.

Shown on graph.

- d. In a symmetrical graph, the mean is the preferred measure of center to use. The histogram for points scored is skewed. Discuss with your partner why the mean would not be a good representation of the center for points scored. Summarize your discussion.

Answers may vary. The value of the mean is affected by the higher values that are causing the shape to be skewed. Using the mean may falsely represent the center as higher than it actually is.

- e. In this graph, the median is located between 15 – 20 and the mean is in the 20 – 25 interval. Which measure most accurately describes the center of the histogram?

The median because the peak and center of the data is closer to 15-20.

- f. Which measure of variation should be used to describe the distribution?

Answers will vary. The range is affected by the few large values that are causing the distribution to be skewed to the right, but they do not affect the IQR. The IQR should be used to describe the variation.



In a skewed graph or a graph with outliers, the **median** is the preferred measure of center and the **IQR** is the preferred measure of variation to describe the distribution.



For a digital learning extension, consider allowing students to enter each number into an Excel spreadsheet and create a chart with the data. Students can then find the mean and median of the given data using the equation functions in Excel.



Students may not understand why the mean would not be a good representation of the center in a skewed graph. To further scaffold this section, ask guiding questions if needed:

- What do you know about outliers? How do they impact data distributions?
- Is the mean impacted by high/low outliers? How?
- Is the median impacted by outliers? Why or why not?



Note that the values of the IQR and range cannot be determined since only the histogram is given. A box plot or data set would need to be included to determine those exact values.

10.2.6 Your Turn!

Component Overview: Students explain which value in a data set they think could be an outlier before describing the effect of removing it from the data set.



10.2.6 Your Turn!

The following data set shows the number of tickets a movie theater sold in a day for each movie.

162	623	76	277	181	64	105
265	82	263	124	167	150	245

- One of the values in the data set is an outlier. Explain which value you think is the outlier.

Answers will vary. 623

This value is much larger than the rest of the data set.

- Complete the statement.

Removing the outlier, 623, causes the mean

to
☐ increase
☒ decrease
☐ remain the same
 and the range to
☐ increase
☒ decrease
☐ remain the same

 because it is much
☒ larger
☐ smaller
 than the other values in the data set.



To challenge students and further build conceptual understanding, ask questions like:

- What are some reasons statisticians may remove an outlier?*
- What are some possible reasons outliers could have occurred in this context?*

10.2.7 Wrap-Up: Cool Down

Component Overview: This Wrap-Up assesses students' abilities to determine if and how an outlier affects different measures of center and variation for a set of data.



10.2.7 Wrap-Up: Cool Down

1. The presence of an outlier has a significant effect on which measures of center and spread? Select all that apply.

- ☒ Mean
- ☐ Median
- ☐ Interquartile Range
- ☒ Range

2. A data set has an outlier that is significantly less than most of the other values.

- a. Explain how this impacts the measures selected in number 1.

A lower outlier will cause the mean to be smaller than the median and will increase the range of the data.

- b. Which measures of center and variation would you use to best describe the data?

Answers may vary. Median and IQR (interquartile range).



Consider using all or parts of this exercise as a formative assessment to gather data on student progress from this lesson and inform groupings for remediation or partner work in the next lesson.

10.3 Justifying Measures of Center and Variation

Lesson Introduction

 Benchmarks	MA.6.DP.1.6 Given a real-world scenario, determine and describe how changes in data values impact measures of center and variation. MA.7.DP.1.1 Determine an appropriate measure of center or measure of variation to summarize numerical data, represented numerically or graphically, taking into consideration the context and any outliers.		 Learning Target	I can justify whether or not a measure of center or variation appropriately summarizes data.
 Benchmark Coherence	Building On MA.5.DP.1.2 Interpret numerical data with whole-number values, represented with tables or line plots, by determining the mean, mode, median, or range.		Working Toward MA.912.DP.1.1 Given a set of data, select an appropriate method to represent the data, depending on whether it is numerical or categorical data, and on whether it is univariate or bivariate. MA.912.DP.1.2 Interpret data distributions represented in various ways. State whether the data is numerical or categorical, whether it is univariate or bivariate, and interpret the different components and quantities in the display.	
 Essential Questions	- How do you determine which measures of center best describe a data set? - How do you determine which measures of variation best describe a data set?			
 Lesson Narrative	In this lesson, students build on their understanding of the effect of outliers from Lesson 2. Students begin by analyzing what information can be determined from different data displays. Students then draw conclusions about how outliers or the shape of a distribution impact the mean, median, range, and IQR and apply this knowledge throughout the rest of the lesson to determine which measures are best to describe data sets given the context of the data presented.			
 Component Summary	10.3.1 Warm-Up (~5 min.)	Notice/Wonder about data in different formats (<i>SE p. 116</i>)	10.3.3 Guided Instruction (10 min.)	Intro to stem-and-leaf plots (<i>SE p. 118</i>)
	10.3.2 Collaboration (10-12 min.)	Review measures of center, variation, and types of data displays (<i>SE p. 117</i>)	10.3.4 Collaboration (15-18 min.)	Describe and interpret measures of center and spread from data displayed in histograms and stem-and-leaf plots (<i>SE p. 119</i>)
	10.3.5 Reflection (~5 min.)	Self-assessment of confidence in the day's learning targets (<i>SE p. 122</i>)		
 Resources	Glossary		Materials	
	<u>Key Terms:</u> - Stem-and-Leaf Plot <u>Additional Terms:</u> - Mean - Range - Median - Interquartile range		Calculators	
			Additional Preparation N/A	

 Teacher Tips	Common Misconceptions - Students may struggle to estimate the mean and median and determine which will be larger based on the shape of a distribution. - Students may not understand which measure of center or variation best describes a given data set. - Students may struggle to connect context to the measures of center and variation.	Things to Consider Consider encouraging students to refer back to the graphs and data displays to help visualize the data. Considering the shape of the distribution is helpful when determining the most appropriate measures of center and spread.
	Literacy and Language Routines Think-Pair-Share 10.3.4 Collaboration provides an opportunity to engage students in a Think-Pair-Share routine as they analyze student responses to data. Have each student think individually about the problem, then discuss it with their partner. As students talk, circulate to find 1-2 responses to share with the class that will advance student thinking and learning.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.7.1: Real-World Contexts This lesson provides a multitude of opportunities for students to engage in discourse that connects mathematics to real-world contexts. Students consider how the context of the data set impacts which measures of center and variation are most appropriate to describe the data.
	 Differentiate Instruction Throughout this lesson, visual entry points are provided by displaying data using various displays such as line plots and histograms. Auditory entry points can be provided through partner and class discussions.	
Lesson Notes:		

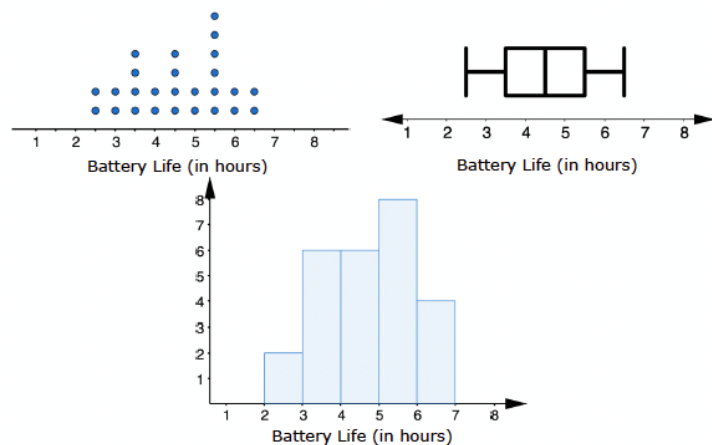
10.3.1 Warm-Up: Notice and Wonder

Component Overview: In this Warm-Up, students will discuss what they notice and wonder about the data shown in multiple formats.



10.3.1 Warm-Up: Notice and Wonder

The dot plot, histogram, and box plot summarize the hours of battery life for 26 cell phones that are constantly streaming videos.



Notice and Wonder

Engage students in a Notice and Wonder routine.

Encourage students to record similarities and differences they notice between the line plot, histogram, and box plot. Ask students what this makes them wonder about the data set or the data displays.

1. What do you notice?

Answers will vary. The dot plot and the histogram look the most similar. The median of the box plot looks to be very close to the middle.

2. What do you wonder?

Answers will vary. I wonder if all the batteries are the same. I wonder if the age of the phone affects the battery life.

10.3.2 Collaboration: Comparing Data Displays

Component Overview: Students revisit the different data displays for numerical data previously learned as well as the concept of an outlier.



10.3.2 Collaboration: Comparing Data Displays

Suppose an office tracks the number of phone calls received on business days, Monday through Friday, over two weeks. The data set is shown below.

183	165	58	152	179
152	153	135	124	112

- Discuss with your partner what you notice about the data set.

Answers may vary. There are two 152's. Most numbers are in the hundreds. Fifty-eight looks out of place.

- Match the measures of center and variation with their calculations for the data set above.

A. Mean	B. Median	C. Range	D. IQR
---------	-----------	----------	--------

B	$\frac{152+152}{2} = 152$
D	$165 - 124 = 41$
A	$\frac{183 + 165 + 58 + 152 + 179 + 152 + 153 + 135 + 124 + 112}{10} = 141.3$
C	$183 - 58 = 125$



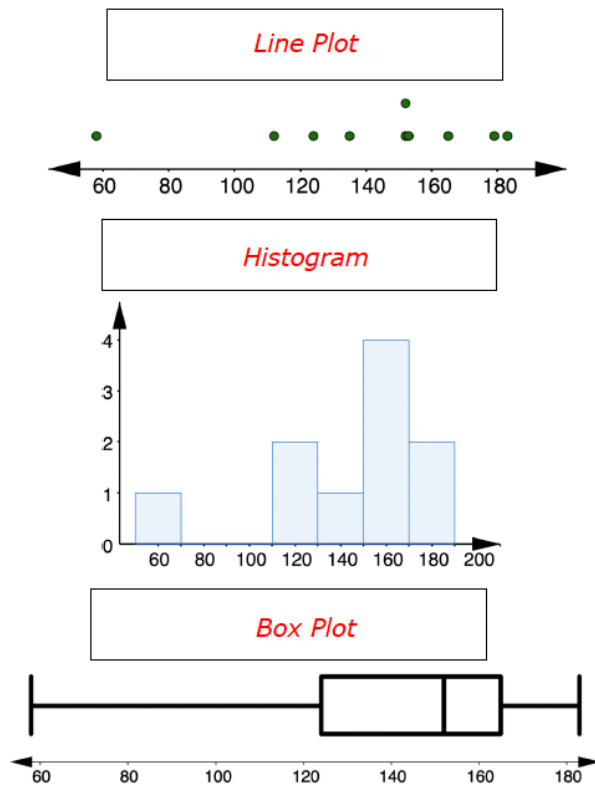
Notice and Wonder

To position students to begin to conceptually consider the concept of outliers, which will be reviewed at the end of this component, engage students in a Notice and Wonder routine. Ask questions like:

- What do you notice about the data set?
- What do the numbers have in common?
- What is different?
- What does this make you wonder?

10.3.2 Collaboration: Comparing Data Displays (continued)

3. A **histogram**, **line plot**, and **box plot** representing the data are shown.



Students represented data using these graphs in Unit 9. Activate prior knowledge through questioning:

- What are the components of a line plot? Histogram? Box plot?
- What do you remember about each of these data displays?
- Which measures of center and variation can be determined from each display?

- a. Label the display types histogram, box plot, or line plot by filling in the box above each display.

10.3.2 Collaboration: Comparing Data Displays (continued)

- b. Complete the table by identifying which measures of center and variation can be determined from each display.

	Mean	Median	Range	IQR
Histogram	☐	☐	☐	☐
Line Plot	☐	☐	☐	☐
Box Plot	☐	☐	☐	☐

4. Describe any outliers in the data set.
58 is an outlier because it is far away from the rest of the values.



5. Explain which of the displays you used to determine whether or not there was an outlier in the data set.
Answers will vary. I used the line plot because I could see that this data point was far away from the other points in the data set. I used the box plot to see that the lowest data point was far from the median.

Students do not need to know how to determine whether a number is an outlier in the Grade 6 Accelerated course, so the values used as potential outliers have already been confirmed in these lessons.

Consider explaining to students that there is a calculation to determine if a value in a data set is actually an outlier. They should not just assume any number larger or smaller than the others in a data set is automatically an outlier, though the examples they'll see in this unit are.

For the teacher's reference, a number is an outlier if it is $1.5 \times \text{IQR}$ above $Q3$ or below $Q1$ in a set of data.

10.3.3 Guided Instruction: Stem-and-Leaf Plots

Component Overview: Introduce students to stem-and-leaf plots, another way to organize and display numerical data.



10.3.3 Guided Instruction: Stem-and-Leaf Plots

Recall the data collected on the number of phone calls an office received on business days over two weeks.

183	165	58	152	179
152	153	135	124	112

In addition to displaying the data graphically with a histogram, line plot, or box plot, the data can also be displayed in a **stem-and-leaf** plot.



A **stem-and-leaf plot** is a table that organizes data by place value to compare data frequencies.

1. What do you notice about the data in the stem-and-leaf plot?

Answers will vary. There is a large gap in the data. There is a number that repeats, 152.

Stem	Leaf
5	8
6	
7	
8	
9	
10	
11	2
12	4
13	5
14	
15	2, 2, 3
16	5
17	9
18	3
Key: 5 8 = 58	



Notice and Wonder

Review the definition of stem-and-leaf plot with students, then use the plot in question 1 to draw students' attention to the features of stem-and-leaf plots. Engage students in a Notice and Wonder routine and circulate as they discuss in groups or pairs.

Students may connect the stem-and-leaf plot to a vertical number line where each ten, in this case, is marked, but there are only values in the second column when there is an actual data value from the set that falls within that interval of 10. Accounting for each interval of 10 within the range allows the display to show the spread of the data.



Stem-and-leaf plots are a useful tool. Students can visually see the large gap in the data, meaning outliers can be more easily identified. Consider creating an anchor chart with students that identifies the parts of stem-and-leaf plots for students to reference throughout this unit.

10.3.3 Guided Instruction: Stem-and-Leaf Plots (continued)

2. Compare the stem-and-leaf plot with the line plot from the collaboration.

What similarities do you notice between the displays?	What differences do you notice between the displays?
<i>Answers will vary. Both displays show a gap in the data.</i>	<i>Answers will vary. The stem-and-leaf plot shows exact values. The line plot's scale makes it hard to determine what the value of each data point is from the display.</i>

3. What value does the entry 13|5 represent? **135**

4. Which measure(s) of center can be determined from the stem-and-leaf plot?

- ☐ mean
- ☐ median
- ☒ both the mean and median

5. Which measure(s) of variation can be determined from the stem-and-leaf plot?

- ☐ range
- ☐ interquartile range
- ☒ both the range and the interquartile range



Display both plots for students to see. Have students work with a partner to discuss the similarities and differences between the displays. As students compare the displays, circulate and ask questions like:

- What do you notice about the values on each display? What does this mean for determining the measures of center and variation?
- What do you notice about the scale of the line plot? Why is this important?



What does it mean when there are no values in the leaf column for some of the stems?

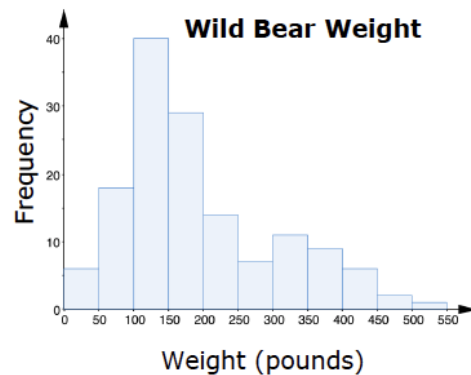
10.3.4 Collaboration: Center and Variation in Context

Component Overview: Students work with a partner to analyze histograms and stem-and-leaf plots. Students use the data displays to answer questions about the data and draw conclusions about the measures of center and measures of variation that best describe each distribution in context.



10.3.4 Collaboration: Center and Variation in Context

1. In a study of wild bears, researchers measured the weights, in pounds (lb), of 143 wild bears that ranged in age from newborn to 15 years old. The data was used to make the histogram shown.



- a. Complete the statements.

Based on the data, it appears wild bears typically

weigh between
☐ 50 and 150
☒ 100 and 200
☐ 150 and 250
 pounds. A good

estimate for the typical weight for wild bears is 175

pounds.



Students may struggle to understand what a “typical weight” is in this context. Consider asking guiding questions like:

- Between what numbers do most of the weights fall in the data set?
- What does typical mean in this context? Is it referring to the center or variation of the data set?
- If the most weights are between 100-150 and 150-200, how can you describe the typical weights of the wild bears?

10.3.4 Collaboration: Center and Variation in Context (continued)

- b. Compare your estimate with your partner. Discuss whether it is possible for both of your estimates to be reasonable, even if they are not the same.
- c. Discuss with your partner whether your estimate from Part A is closest to the median or the mean. Summarize your discussion.
Answers will vary. The estimate would be closest to the median. The few bears in the data set that weigh more will cause the mean to be higher than the median.
- d. Angelica and Raymond were asked to describe the variation in weight of the wild bears.

Angelica said, "The bear weights range from less than 50 to over 500 lbs, so the range best describes the spread of the weights of the wild bears."

Raymond said, "It looks like most bears in the sample weigh between 100 and 300 lbs, with few weighing less and few weighing more. The small weights could be because there were babies in the sample. So, the interquartile range is the best measure of spread to describe this data."

Explain whether you agree with Angelica or Raymond. Justify your reasoning.

Answers will vary. Raymond's explanation best describes the spread. He justifies his choice with the understanding that the small weights are because the bears were babies and the frequency for the weights between 100 and 300 is much higher than at heavier weights.



Auditory entry points are built in throughout this component through partner discussion. Consider providing additional visual entry points through the use of diagrams. Encourage students to use colored pencils to mark the median and mean on the histogram to help visualize their estimations and response.

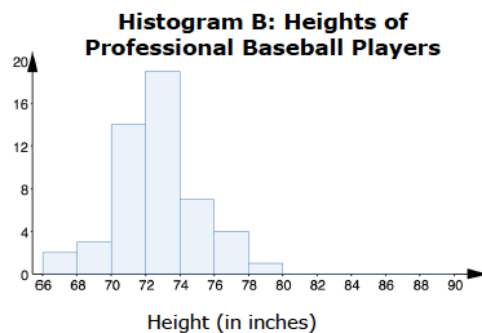
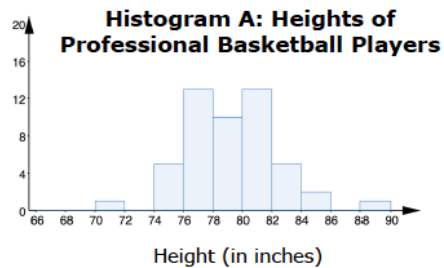


Think-Pair-Share

Consider implementing a Think-Pair-Share routine for question 1d. Have each student think individually about the problem, then discuss it with their partner. As students talk, circulate to find 1-2 responses to share with the class that will advance student thinking and learning. Listen for students who can justify why the IQR makes the most sense based on the shape of the distribution.

10.3.4 Collaboration: Center and Variation in Context (continued)

2. The two histograms, labeled A and B, show height distributions of 50 male professional basketball players and 50 male professional baseball players.



Work with your partner to complete each statement.

- a. In ☐ Histogram A, ☒ Histogram B, the average height is around 72 inches. The values of the mean and median are similar.
- b. To explain typical height, the variation in ☒ Histogram A ☐ Histogram B is best described by the interquartile range because the range may be misleading due to outliers. The values of the mean and median are similar.



Students may easily guess the answers to question 2. However, connect it to the next activity by asking questions like:

- What values appear to be outliers?
- What would the mean and median be of each histogram?
- Is the mode easiest to find when given a histogram?
- On average are basketball players or baseball players taller?

10.3.4 Collaboration: Center and Variation in Context (continued)

3. An earthworm farmer set up several containers of a certain species of earthworm so he could learn about their lengths. Lengths of earthworms provide information about their ages. The farmer measured the lengths of the earthworms in one container.

The farmer displayed the earthworms' lengths, in millimeters (mm), in a stem-and-leaf plot.

Stem	Leaf
0	6
1	1, 8, 9
2	0, 3, 3, 5, 5, 6, 7, 7, 8, 9
3	2, 3
4	1, 2, 8
5	2, 4, 9
6	0
7	7
8	
9	3

Key: $2|8 = 28$

Use the display to answer the following questions.

- Half of the earthworms sampled were less than **28** mm long.
- The farmer reported that the average length is 35.92 mm. Why might this measure of center be misleading about the typical length of the earthworms in this sample?
Answers will vary. This could be misleading because of the vast range of data.
- Compare your answers to Parts A and B with your partner, and if necessary, resolve any differences.



If students are struggling to answer question 3b, provide additional scaffolding through intentional questions like:

- What do you notice about the spread of the data? Why is this important when considering measure of center?
- Based on what you know so far about measures of center, how can outliers impact the mean, or average?
- Is there one measure of center that would be more accurate to describe the data? Why?

10.3.4 Collaboration: Center and Variation in Context (continued)

- d. The farmer reported that the lengths varied between 6 mm and 93 mm. The farmer's assistant said that most earthworms had lengths between 23 and 50 mm.

Work with your partner to complete the statements.

The farmer is using the ☐ IQR
☒ range to summarize

the variation in worm length, while the farmer's

assistant is using the ☒ IQR.
☐ range.

If a researcher is interested in how long

earthworms are, the ☒ IQR
☐ range is an appropriate

measure of variation for the farmer to report

because *Answers will vary. IQR is not affected by outliers.*



While students are working, consider any observational data based on student responses and discussions when completing 10.3.4 Collaboration. Use the observations to inform what differentiation or small-group instruction can look like in future lessons to ensure students are able to correctly determine which measures of center and variation to use to describe a data set as well as how to interpret those measures in context.

10.3.5 Wrap-Up: Reflection

Component Overview: Students complete a self-reflection to indicate their current understanding of the lesson's learning targets.



10.3.5 Wrap-Up: Reflection

Draw an **X** on the line to indicate where your current learning is in relation to the following learning targets.

1. I can justify whether or not a measure of center appropriately summarizes data.

Answers will vary.

I need help.
I can't get started.

I'm getting there, but
I need more practice.

I understand this
and I feel confident.



The information provided from the student here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.

2. I can justify whether or not a measure of variation appropriately summarizes data.

Answers will vary.

I need help.
I can't get started.

I'm getting there, but
I need more practice.

I understand this
and I feel confident.



10.4 Determining Appropriate Measures of Center and Variation

Lesson Introduction

 Benchmark	MA.7.DP.1.1 Determine an appropriate measure of center or measure of variation to summarize numerical data, represented numerically or graphically, while taking into consideration the context and any outliers.		 Learning Target	I can determine an appropriate measure of center and variation to summarize numerical data.
 Benchmark Coherence	Building On MA.6.DP.1.2 Given a numerical data set within a real-world context, find and interpret mean, median, mode, and range. MA.6.DP.1.6 Given a real-world scenario, determine and describe how changes in data values impact measures of center and variation.		Working Toward MA.912.DP.1.1 Given a set of data, select an appropriate method to represent the data, depending on whether it is numerical or categorical data and on whether it is univariate or bivariate.	
 Essential Questions	- How do you determine which measures of center best describe a data set? - How do you determine which measures of variation best describe a data set?			
 Lesson Narrative	In this lesson, students continue to build their understanding of how to determine which measures of center and variation best describe sets of data based on real-world contexts. Emphasis should be placed on the context, not only the visual representation of the data. In the Guided Practice, students work through scaffolded prompts to determine the measures of center and variation that most accurately describe the given data sets. As the lesson progresses, students continue to interpret data sets and describe them using appropriate measures of center and variation through less scaffolded means, culminating in a Wrap-Up requiring students to interpret and describe data presented in a line plot.			
 Component Summary	10.4.1 Warm-Up (~5 min.)	Create a ratio to represent a real-world context (<i>SE p. 123</i>)	10.4.3 Try It (10-15 min.)	Explain which measures of center and variation most accurately synthesize a data set (<i>SE p. 125</i>)
	10.4.2 Guided Practice (10-15 min.)	Describe measures of center and variation to best summarize a data set (<i>SE p. 123</i>)	10.4.4 Your Turn! (10-15 min.)	Interpret a histogram to determine the appropriate measures of center and variation to describe the data (<i>SE p. 126</i>)
	10.4.5 Wrap-Up (~5 min.)	Examine the data in a line plot to choose the best measures of center and variation and explain why these measures most accurately describe the data (<i>SE p. 128</i>)		
 Resources	Glossary	Materials		Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Mean - Median - IQR - Range	N/A		N/A

 Teacher Tips	Common Misconceptions	Things to Consider
	Students may forget which measures of center and variability are best for data sets with outliers or skews.	Encourage students to refer back to the summary they previously completed when considering which measures of center and variation to use to summarize data sets.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Notice and Wonder For the 10.4.3 Try It, consider implementing a Notice and Wonder routine before students answer question 1. As students discuss what they notice about the data, circulate to listen for students who are noticing the outliers in the data and discussing what that means for the measures of center and variation. Highlight these for the class in a whole-group discussion.	MA.K12.MTR.7.1: Real-World Contexts Throughout this lesson, students connect measures of center and variation to the contexts of the data sets and their purpose in the real world. Students use context to determine which measures are most appropriate to summarize and make decisions about the data set.
 Differentiate Instruction	10.4.2 Guided Practice offers opportunities to include multiple entry points for students to access the content. To provide auditory entry points, students can work in pairs or small groups and discuss each problem. Visual entry points can be accessed through diagrams representing the data. Encourage students to draw box plots, histograms, or stem-and-leaf plots to help visualize the data before answering the questions.	
	Consider adding scaffolding questions to the 10.4.4 Your Turn!, similar to what is included in the 10.4.2 Guided Practice, to help students who are struggling to determine which measures of center and variation best describe the data sets. This adjustment will provide all students with an entry point to be able to apply their learning to drive conceptual understanding.	
Lesson Notes:		

10.4.1 Warm-Up: Between Two Numbers

Component Overview: Students work independently to create a ratio that represents the ratio of the sun's core temperature to the Earth's core temperature.



10.4.1 Warm-Up: Between Two Numbers

The temperature of the sun's core is 15 million degrees Celsius.

The temperature of Earth's core is 6000 degrees Celsius.

1. What is the ratio of core temperatures of the sun to Earth?

$$\frac{15,000,000}{6,000} = \frac{15,000}{6} = \frac{2,500}{1}$$



This Warm-Up reminds students of one of the first methods they used to compare just two numbers: ratios. Throughout this unit students will be asked to make comparisons within sets of data.

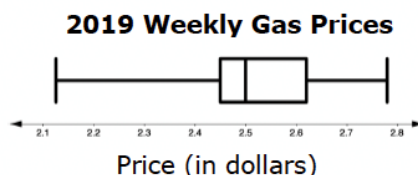
10.4.2 Guided Practice: Measures of Center and Variation

Component Overview: Students determine and explain which measures of center and variation best describe data displayed in a box plot and stem-and-leaf plot based on the context and purpose.



10.4.2 Guided Practice: Measures of Center and Variation

- The U.S. Energy Information Administration reports weekly average gas prices for the United States. The box plot represents the weekly average gas prices in 2019.



- Explain which measure of center, the mean or the median, best describes the center of the data.
The median best describes the center of data. The first quartile is much larger than the others, suggesting that the data is skewed left.

- If possible, find the value of the measure you chose in Part A.
The median is 2.5 which is \$2.50 per gallon.

- Describe the variation of the weekly average price of gasoline.
Answers will vary. The middle 50% of data values, the IQR, are relatively close together. In the first quartile of data, values are more spread out. Variation is described in terms of range and interquartile range (IQR). Because the box plot does not give exact numbers for the lower and upper extreme or the quartiles, students may approximate the range to be around \$0.70 and the IQR to approximately \$0.20.

- Given your description, does the interquartile range or range better represent the variation in price?
Answers will vary. Because the box plot is skewed left, the IQR better represents the variation in price.



Students may still struggle to conclude which measures of center/variation best describes the data. To scaffold, consider asking questions like:

- Does this data set look like it is skewed or could have outliers? Why?
- When outliers are present or the distribution is skewed, which measure of center/variation usually best describes the data? Why?

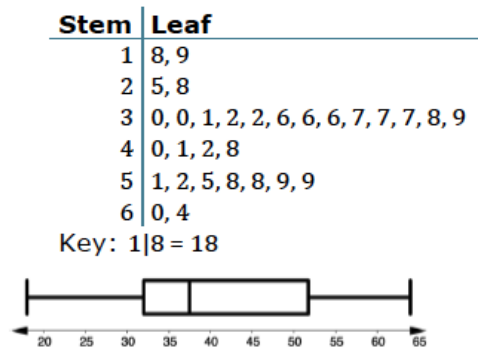


Consider providing sentence starters to provide additional support for students who may struggle to explain their thinking in writing.

- "Variation in a data set is described using _____ and _____. The _____ of the data is around _____. The _____ is approximately _____."
- "The _____ would better represent the variation in price."

10.4.2 Guided Practice: Measures of Center and Variation (continued)

2. Robin, a small business owner, is trying to determine the cost of providing a small group health care plan to her 30 employees. Her insurance agent asks her for a summary of the employees' ages, since age is a factor that affects the premium cost of insurance. The employees' ages are displayed in the stem-and-leaf plot and box plot.



Employee Ages (in years)

Complete the statements to explain which measure of center and variation Robin should provide to her insurance agent to accurately summarize the ages of her employees.

- a. The ☐ mean ☐ median best summarizes the center of

the data for the agent because ...

Answers will vary. The data is approximately symmetric, so the mean and median values are close. Generally, in this case, the agent would use the mean.

- b. The ☐ IQR ☐ range best summarizes the variation of

the data for the agent because ...

Answers will vary. The range should be provided since the cost of insurance depends on the ages. If only the IQR is provided, the cost may be underestimated.



Students may immediately choose IQR as the best measure of variation since they are shown a box plot. It is important to emphasize how the context of the data should inform which measure to use. In this case, the cost of the insurance premium is affected by age, so the high and low values are important to consider; thus, the range is a more helpful measure of variation to accurately estimate the insurance premium.

10.4.3 Try It: Measures of Center and Variation

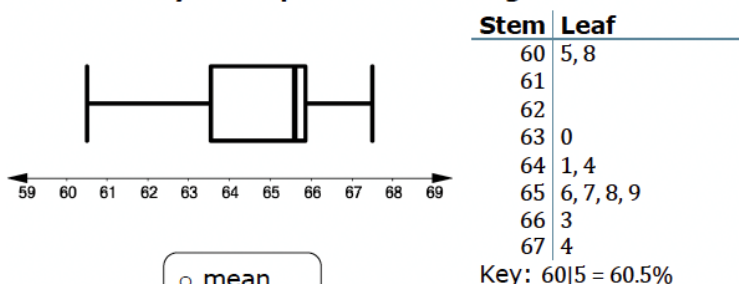
Component Overview: Students work independently or with a partner to analyze data displayed in a box plot and stem-and-leaf plot, completing statements to describe the data. Then students evaluate the data of home prices to determine which measures of center and variation best summarize the average home price in a neighborhood.



10.4.3 Try It: Measures of Center and Variation

1. Tom Brady is considered by most to be one of the best NFL quarterbacks of all time. His pass-completion percentages from 2009 to 2019 are shown in the box plot and stem-and-leaf plot.

Tom Brady's Completion Percentages 2009-2019



- a. The ☐ mean ☒ median best summarizes the center of the data. The ☒ mean ☐ median is not the best measure of center because it is influenced by the values ☒ 60.5% and 60.8% ☐ 66.3% and 67.4% that are far from the rest of the data values.

- b. The ☒ IQR ☐ range best summarizes the variation of the data. The ☐ IQR ☒ range is not the best measure of variation because it is affected by the value ☒ 60.5% ☐ 67.4% that is far from the rest of the data.



Notice and Wonder

It may be helpful to implement a Notice and Wonder routine before students complete question 1. This will provide multiple entry points through low stakes questioning, ensuring all students can access the material in some way. As students are discussing what they notice about the data, circulate to listen for students who are noticing the outliers in the data and discussing what that means for the measures of center and variation used to describe the distribution. Highlight these for the class in a whole-group discussion.

10.4.3 Try It: Measures of Center and Variation (continued)

2. The Rachards family is moving to Gainesville, FL and want to purchase a home in a specific area. They have a budget of \$160,000. The prices of homes in the area are shown below, in thousands of dollars.

150, 302, 201, 212, 150, 160, 140, 145

The Rachards are concerned that the homes may be out of their budget, so they ask the realtor to summarize the home prices.

- a. Which measure of center should the realtor report to the Rachards, the mean or the median? Explain your reasoning.

Answers will vary. The realtor should report the median to the Rachards because it is not affected by the extreme value of 302 thousand dollars. If the realtor gives the mean, the Rachards may think they can't afford a home in the area.

- b. Which measure of variation should the realtor report to the Rachards, the interquartile range or the range? Explain your reasoning.

Answers will vary. The realtor should report the interquartile range to the Rachards because it provides the variation of the center of the data.



Consider ways to provide multiple entry points for students to access the content in this component. To deliver auditory entry points, students can work in pairs or small groups and discuss each problem. Visual entry points can be accessed through diagrams representing the data. Encourage students to draw box plots, histograms, or stem-and-leaf plots to help visualize the data before answering the questions.

10.4.4 Your Turn!

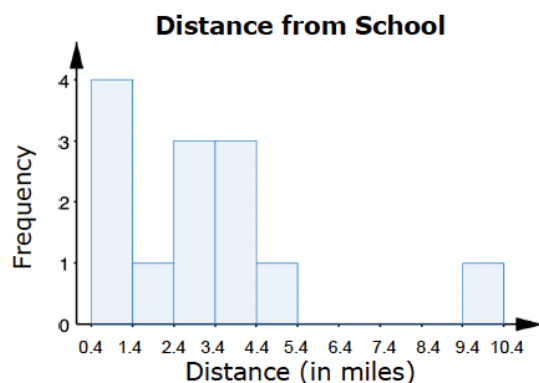
Component Overview: Students work independently to analyze data displayed in a histogram, determining which measures of center and variation best describe the data and why. Students then consider whether the measures of center and variation need to change once the outlier of the data set is removed.



10.4.4 Your Turn!

Zane and 12 of his friends listed how far, in miles, they live from the school and created a histogram.

0.4, 0.5, 1, 1, 2.4, 2.7, 3, 3.2, 4, 4.2, 4.4, 5, 10.3



1. Zane concluded that the best description of the typical distance that a student in his class lives from school is 3 miles, the value of the median.

Do you agree with Zane? Explain your reasoning.

Answers will vary. Zane's decision to use the median to describe the typical distance that a student in his class lives from school is the best because it is not affected by the outlier of 10.3.

2. Zane explained to his classmate, Shelby, that the students lived anywhere from 0.4 to 10.3 miles from school, so the range of 9.9 miles was the best summary for the spread of the data.

Shelby argued, "That's misleading, only one person lived more than 5.4 miles from school, so the interquartile range is a more accurate depiction of the spread of the distances."



Students may have trouble explaining their reasoning. If this is the case, you can scaffold the entry point by reminding them to consider the outlier(s). You may, also, encourage them to utilize the units on the horizontal axis as a means to the real-world context of the question.

10.4.4 Your Turn! (continued)

- a. Do you agree with Zane or Shelby? Explain your reasoning.

Answers will vary. Shelby's use of the interquartile range is a more accurate depiction of the spread of the distances because it is not affected by the outlier of 10.3.

- b. Under what circumstances would you agree with the other person's chosen measure of variation?

Answers will vary. If the intent was to estimate how long it would take to visit each person's house, the range would be useful since the high value would impact the time required.

3. Noelle is Zane's friend who used to live 10.3 miles away from the school. She recently moved closer, and now only lives 5.5 miles away from the school.

The data set with Noelle's new distance from school, in green, is shown.

0.4, 0.5, 1, 1, 2.4, 2.7, 3, 3.2, 4, 4.2, 4.4, 5, 5.5

- a. What measure of center best describes the typical distance Zane's friends live from school? Explain your reasoning.

Answers will vary. Because the data are more symmetrical, Zane should use the mean to describe the measure of center.

- b. What measure of variation best describes the spread of the data distribution? Explain your reasoning.

Answers will vary. Since the distribution is more symmetrical, the range is the best measure of variation to describe the spread of the data.



Students may struggle to explain their reasoning in question 2b. If you notice students struggling, ask questions to further scaffold like:

- When does the range/IQR best summarize the data? What does the shape look like? What would the data look like on a box plot or line plot?
- Does the range/IQR best summarize a data set with outlier? Why or why not? What purpose for analyzing the data might change this selection in this context?



Clarifications for this benchmark include recognizing whether a measure of center or measure of variation is appropriate and can be justified based on the given context or the statistical purpose. Throughout this component, students are required to explain and justify their reasoning for their responses.



Challenge students who finish early to represent the data presented using a histogram, line plot, or box plot. Then, have them explain how their representation supports their responses.

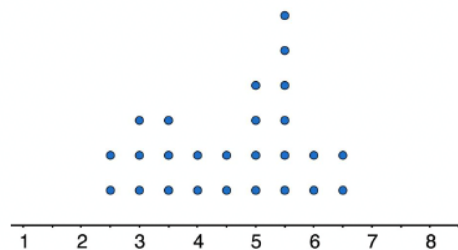
10.4.5 Wrap-Up: Ticket Out the Door

Component Overview: Students examine data in a line plot to determine measures of center and variation and explain which of these measures best describes the data.



10.4.5 Wrap-Up: Ticket Out the Door

The dot plot summarizes the hours of battery life for students' cell phones when constantly streaming videos.



Battery Life (in hours)

Mr. Funk's world history class has an upcoming field trip to Tallahassee, FL that involves a five-hour drive. The students want to be able to watch videos on their phones during the drive without charging their cell phones.

1. Determine the appropriate measure of center and variation for the battery life of the students' cell phones when streaming videos.

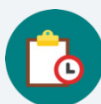
Answers will vary. The time the phones will be able to stream ranges from 2.5 hours to 6.5 hours. The mean is approximately 4.6, the median is 5, the mode is 5.5, the range is 4, and the IQR is 2. Because there is no extreme low or high, the mean most accurately describes the center of the data. The interquartile range best describes the spread of the data. The range is important to consider, however, since some students' phones will not last for the entire five-hour drive while streaming.



The information provided from the student here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.

10.5 Comparing Graphical Representations

Lesson Introduction

 Benchmark	MA.7.DP.1.2 Given two numerical or graphical representations of data, use the measure(s) of center and measure(s) of variability to make comparisons, interpret results, and draw conclusions about the two populations.		 Learning Target	I can use two graphical representations of data to make comparisons and draw conclusions about the populations.
 Benchmark Coherence	Building On		Working Toward	
	MA.6.DP.1.3 Given a box plot within a real-world context, determine the minimum, the lower quartile, the median, the upper quartile and the maximum. Use this summary of the data to describe the spread and distribution of the data. MA.6.DP.1.4 Given a histogram or line plot within a real-world context, qualitatively describe and interpret the spread and distribution of the data, including any symmetry, skewness, gaps, clusters, outliers, and the range.		MA.912.DP.2.1 For two or more sets of numerical univariate data, calculate and compare the appropriate measures of center and measures of variability, accounting for possible effects of outliers. Interpret any notable features of the shape of the data distribution.	
 Essential Questions	- How can you use data displays to compare two data sets? - What measures of center and variation best describe data sets with and without outliers? - How can you use measures of center and variation to draw conclusions about data sets?			
 Lesson Narrative	In this lesson, students apply their knowledge of the measures of center and variation to compare data sets. After a brief Guided Practice, students rotate through three stations comparing data sets presented in line plots, box plots, and stem-and-leaf plots. Students use the data displays to describe the data sets and draw conclusions about the data based on their comparisons.			
 Component Summary	10.5.1 Warm-Up (~5 min.)	Estimate and determine defrosting time using rates (SE p. 129)	10.5.4 Activity: Station 2 (10 min.)	Compare data presented in box plots (SE p. 131)
	10.5.2 Guided Practice (5-10 min.)	Match line plots to data sets based on descriptions (SE p. 129)	10.5.5 Activity: Station 3 (10 min.)	Compare data presented in stem-and-leaf and box plots (SE p. 132)
	10.5.3 Activity: Station 1 (10 min.)	Compare data presented in histograms and line plots (SE p. 130)	10.5.6 Wrap-Up (~5 min.)	Self-reflection on the day's learning target (SE p. 133)
 Resources	Glossary	Materials	Additional Preparation	
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> N/A	(Optional) Warm-Up Video	- Warm-Up 7.10.5 contains an optional video to accompany the question prompts. - Consider using data from the previous lesson's Ticket out the Door to create small groups ahead of time for the station activities (optional)	

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may struggle to draw conclusions from just graphical representations. - Students may struggle to explain the reasons for differences in data for certain contexts. - Students may struggle to recognize that faster means a smaller value in the context of race times.	- Encourage students to consider the features of the data displays, such as shape, spread, and center, to help draw conclusions about the data. - Activate prior knowledge as students work through each scenario to provide additional context for the data. - Consider which groups or stations may need additional support from the teacher and keep this in mind when setting up the flow of each rotation.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Take Turns 10.5.4 Activity: Station 2 provides an opportunity to engage students in a Take Turns routine. Students each participate in the work of the activity, then take turns explaining and justifying their responses. This routine encourages all students to participate and prevents 1 or 2 students from monopolizing the mathematical discussion.	MA.K12.MTR.4.1: Discussion and Reflection Each of the Activity Stations engages students in mathematical discourse through partner and small-group work. Encourage students to clarify, critique, and elaborate on their peers' answers to develop students' ability to justify methods and compare their responses to the responses of their peers.
 Differentiate Instruction	10.5.3, 10.5.4, and 10.5.5 Activities provide multiple entry points for students to engage with the content. The graphical representations of the data sets provide visual comparisons that will be particularly helpful for students struggling to compare the data sets. Each activity station is completed through partner or small-group work, providing auditory entry points as students discuss and reason with their peers to complete each activity.	
Lesson Notes:		

10.5.1 Warm-Up: Defrost Time

Component Overview: Play the “Defrost Black Box” video for the class, then release students to work independently to answer the questions.



10.5.1 Warm-Up: Defrost Time

1. A microwave shows that it takes 9 minutes and 45 seconds to defrost 1.5 pounds of meat and 26 minutes to defrost 4 pounds of meat.
 - a. Without performing any calculations, estimate a range of minutes you think it will take to defrost 3.5 pounds of meat.
Answers will vary. It will take approximately 20 to 24 minutes defrost 3.5 pounds of meat.
 - b. Explain how the given information could be used to find the exact amount of time it takes to defrost 3.5 pounds of meat.
Answers will vary. Using the given rates, find the unit rate (time for one pound). Using the unit rate, scale up for the time it would take to defrost 3.5 pounds.
 - c. Determine how long it would take for the microwave to defrost 3.5 pounds of meat. Show your work.
Answers will vary. It will take 22 minutes and 45 seconds to defrost the meat in the microwave.

$$\frac{26}{4} = \frac{6.5}{1} = \frac{22.75}{3.5}$$



This activity provides visual entry points through the presentation of information in the video.

10.5.2 Guided Practice: Comparing Line Plots

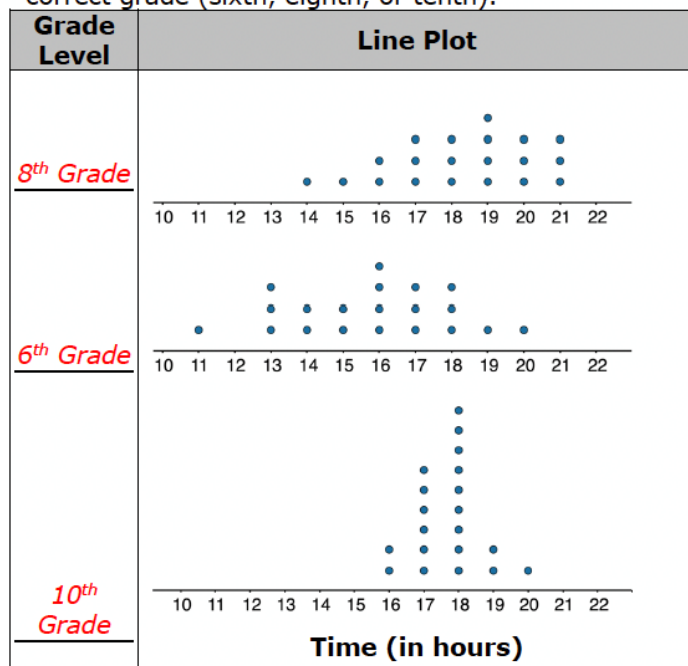
Component Overview: Students match line plots to a data set based on summaries of each set. Students will then rotate through 3 activity stations to continue to apply graphical knowledge of data to make comparisons and draw conclusions about data.



10.5.2 Guided Practice: Comparing Line Plots

- Clare recorded the amount of time a sample of students in the sixth, eighth, and tenth grades spent doing homework, in hours per week. She made a line plot of the data for each grade and provided the following summary:
 - Students in sixth grade tend to spend less time on homework than students in eighth and tenth grades.
 - The homework times for the tenth-grade students are more alike than the homework times for the eighth-grade students.

Use Clare's summary to match each line plot to the correct grade (sixth, eighth, or tenth).



Students completed a similar comparison in Unit 9. The activity reappears as an opportunity to revisit the type of reasoning needed to make the matches and to probe for additional deeper thinking. At this point, students should be able to make additional comparisons between the distributions and discuss more confidently. If, using formative data from prior lessons, you determine that your students do not need this additional support, consider jumping straight into stations following the Warm-Up component.



If students are struggling to make the connection between the line plots and corresponding grade levels, ask prompting questions to provide further scaffolding:

- Where would you expect the 6th grade data to mostly fall on the number line? Why?
- If 10th graders' homework times are "more alike" than 8th graders', does that mean they have more variability or less? How do you know?
- How does a line plot with little variability look compared to one with a lot of variability?



At the conclusion of this component, split students into pairs or small groups and assign each a station to start at. Each group should rotate through all three stations and complete the activities. Allot specific times for each station and review with the class what the signal will be when it is time to transition to the next station.

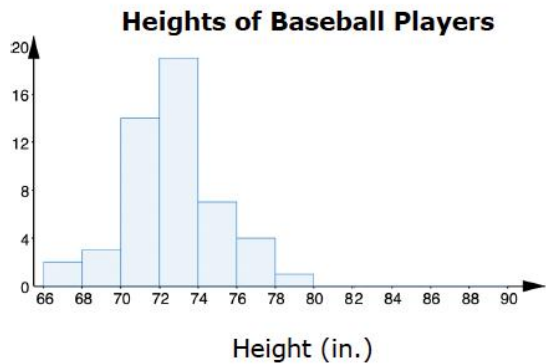
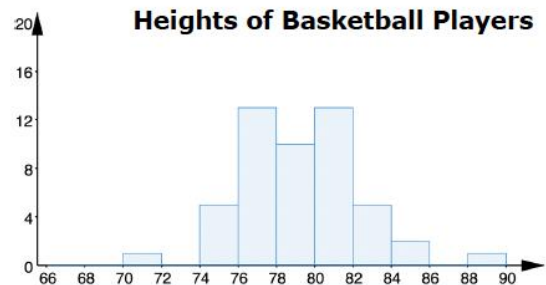
10.5.3 Activity: Station 1 - Histograms and Line Plots

Component Overview: Students compare pairs of data sets presented in histograms and line plots.



10.5.3 Activity: Station 1 – Histograms and Line Plots

- The two histograms show height distributions of 50 male professional baseball players and 50 male professional basketball players, in inches (in.).



This component provides visual entry points through the display of data on the histograms and line plots. Student discussion provides auditory entry points.

10.5.3 Activity: Station 1 - Histograms and Line Plots (continued)

- a. Complete the statements that draw a conclusion about the typical height of professional basketball players and baseball players.

The sample data in the histograms shows the

center of the data for basketball players is

- ☐ less than
☒ greater than

the center of the data for baseball

players. It can be concluded that professional

basketball players are typically

- ☐ shorter than
☒ taller than

professional baseball players.

- b. Complete the statement.

The histograms show that in the samples, the

range of heights of basketball players is

- ☐ less than
☒ greater than

the range of the heights of

baseball players. We can conclude that professional

basketball players have

- ☒ more
☐ less

variability in

their heights than professional baseball players.

One possible reason for this is ...

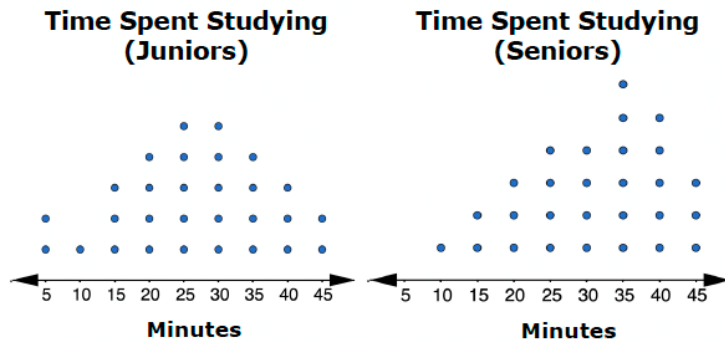
Answers will vary. A possible reason for the variation could be the different heights have benefits for different positions on a basketball team.



Question 1b challenges students to consider why there is a difference in the heights of basketball players vs. baseball players. Students who may not watch or know a lot about sports may struggle to answer this question because they are not familiar with the context. Have group members discuss what they know about the skill requirements for each sport and how that might impact the populations/data.

10.5.3 Activity: Station 1 - Histograms and Line Plots (continued)

2. Mr. Logan surveyed a sample of his junior and senior students about the time they spend studying math per day. He then tabulated the results and created a line plot displaying the data for each group.



- The value of the larger median study time for the two groups is 32.5.
- The value of the larger mean study time for the two groups is 30.9.
- In one to two sentences, describe the difference between the number of minutes Mr. Logan's juniors and seniors study by comparing the center and shapes for the groups.

Answers will vary. According to the mean of both groups, seniors study an average of 4.3 minutes longer than juniors. The median study time for seniors is 7.5 minutes higher than the median for juniors. The juniors' line plot appears more symmetrical than the seniors, so there are some seniors who are pulling the mean hours studying above the median.



As students work through this component in their small groups, circulate and ask guiding questions to prompt their thinking:

- What do you notice about the spread of the data on each line plot? What does this tell you about the time each group spends studying?
- Without doing the calculations, which measures of center do you think are best to describe each data set? Why?

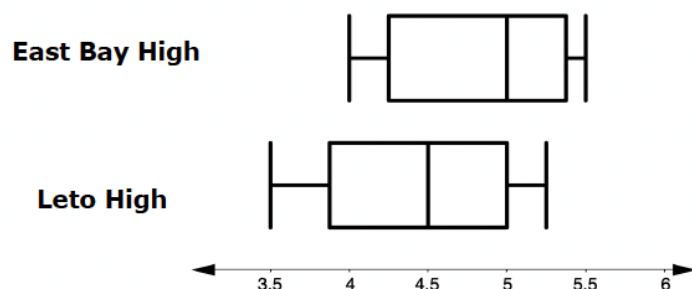
10.5.4 Activity: Station 2 - Box Plots

Component Overview: At Station 2, students compare data presented in two box plots and describe features of the data sets based on the box plots. Students then draw conclusions based on the presented data and what they know about measures of center and variability and conclude by creating a summary describing what they know about the data sets.



10.5.4 Activity: Station 2 – Box Plots

- The mile times for a sample of track team members at East Bay High School and Leto High School are shown in the two box plots.



Mile Time (minutes)

- Using the box plots above, calculate and describe each of the features for Leto High in the table.

Feature	East Bay High School	Leto High School
Shape	The shape of the data is asymmetrical.	<i>Answers will vary. The shape of the data is asymmetrical.</i>
Center	The median mile time for the sample of East Bay High team members is 5 minutes.	<i>The median mile time for the sample of Leto High is 4.5 minutes.</i>
Variability	The range of the sample of East Bay High team members' mile times is 1.5 minutes. The IQR of the sample of East Bay High team members' mile times is 0.75 minutes.	<i>The range of the sample of Leto High team members' mile times is 1.75 minutes. The IQR of the sample of Leto High team members' mile times is 1.2 minutes.</i>



Take Turns

Consider implementing a Take Turns routine during this station. Have pairs or groups work to complete each part of this activity, then take turns justifying their responses to their peers. Encourage students to explain whether they agree or disagree with each other's' arguments and why.

10.5.4 Activity: Station 2 - Box Plots (continued)

- b. Based on the data, explain which team is more likely to win at a track meet.
Answers will vary. Leto High would be more likely to win at a track meet. Their median time is faster than East Bay.
- c. Suppose a runner at a track meet runs a mile in 3 minutes and 45 seconds. Explain which track team you would assume they were on, using information from the data displays.
Answers will vary. The runner is most likely a member of the Leto High team because their data supports a runner finishing under 4 minutes.
- d. Based on your conclusions and answers to Parts A through C, use some of the terms from the word bank below to write a 4-5 sentence summary comparing the population of runners at East Bay High School to the population of runners at Leto High School.

mean	median	range	IQR
center	variability	upper quartile	
lower quartile	sample	population	

Answers will vary. Leto High School has faster runners than East Bay High. The median time for Leto High is 4.5 minutes where East Bay has a median of 5 minutes. On average, Leto High is finishing half a minute faster than East Bay. East Bay High's spread is smaller than Leto High, but they have times longer than Leto High.



Students may struggle to recognize that faster times mean a shorter completion time. For example, some students may think East Bay High is faster because their median is higher.



As students work through this station, ask questions to encourage synthesis of the material:

- How can you use the box plots to determine which team is more likely to win the track meet?*
- Which team is faster? How do you know? How can you use this to help you answer the questions and summarize the data?*



Challenge students who may finish early to make predictions about how different times might impact the spread of the data sets. What if the runner who ran a mile in 3 minute and 45 seconds was a member of the East Bay High School team? Would this significantly impact the data? Why or why not?



Students may have a hard time drawing conclusions about the populations to compare the data. Remind them to compare shape, center, and variability. Ensure students understand that the data sets are generally representative of the populations and can therefore be predictive of the populations at each school.

10.5.5 Activity: Station 3 - Stem-and-Leaf Plots and Box Plots

Component Overview: At Station 3, students analyze the data of two groups displayed in stem-and-leaf plots and two groups displayed in box plots. Students draw conclusions about the data sets and apply their knowledge of measures of center and variability to make determinations about the data.



10.5.5 Activity: Station 3 – Stem-and-Leaf Plots and Box Plots

1. A sample of two stores' guitar prices are shown in the stem-and-leaf plots.

Store A's Guitar Prices

Stem	Leaf
2	1, 2, 8
3	3, 6
4	5, 8, 8
5	1, 6
6	
7	2, 2
8	1, 6
9	1

Key: 2|1 = \$210.00

Store B's Guitar Prices

Stem	Leaf
4	5, 8
5	6, 9
6	1, 1, 3, 5
7	5, 5, 6
8	0, 6
9	1, 8
10	0

Key: 2|1 = \$210.00

Zach wants to purchase a guitar and is trying to determine which store to purchase from. Using the displays he makes the following conclusion.

"Store B is better for more experienced guitar players because it typically sells more expensive guitars than Store A. The median price of Store B's guitars is \$700, while the median price of Store A's guitars is \$480. Prices of guitars at Store A tend to vary more, with an IQR of \$390, compared to the prices of guitars at Store B, with a much smaller IQR."

Do you agree or disagree with Zach's conclusion? Explain your reasoning.

Answers will vary. The calculations Zach has made are correct. However, those calculations alone cannot determine if Store B is for more experienced guitar players. Store B could just carry higher quality or specific name brands of guitars whereas Store A may sell a variety of name brands and quality.

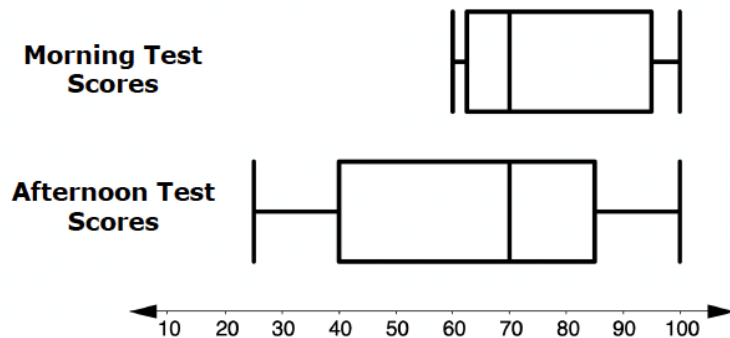


Think-Pair-Share

Engage students in a Think-Pair-Share routine as they determine whether they agree or disagree with Zach's conclusion. First, students analyze the stem-and-leaf plots and consider whether they agree with Zach's response. Next, students share their thoughts with a partner. Encourage students to explain whether they agree or disagree with their partner and why. Finally, students complete their written response that may later be shared with the class.

10.5.5 Activity: Station 3 - Stem-and-Leaf Plots and Box Plots (continued)

2. Suppose a sample of test scores is taken from Mr. Spellman's morning math classes. The data is displayed in the top box plot. Another sample of test scores is taken from Mr. Spellman's afternoon math classes. The data is displayed in the bottom box plot.



Explain the difference between taking Mr. Spellman's math test in the morning compared to the afternoon. Use some of the terms from the word bank and be sure to include both a measure of center and variability in your explanation.

mean	median	range	IQR
center	variability	upper quartile	
lower quartile	sample	population	

Answers will vary. The mean cannot be determined from the box plots above. The median would be the best measure of center. Both the morning and afternoon students had a median score of 70. The variation in scores for the morning students is lower than the afternoon students. The range for the morning students was 40 whereas the range for the afternoon students was 75. There are a variety of factors that could cause the spread in data to be different such as students did not study or they were more alert in the morning as compared to the afternoon.



As students complete this component, ask questions to solidify understanding:

- Which measures of center/variability are best to use for the data in the box plots? Why?
- How do the representations look different?
- What do you notice about the box plots just by looking at them?
- What conclusions can you make about the samples?

10.5.6 Wrap-Up: Reflection

Component Overview: Students complete a reflection to determine what they are still unsure of after the lesson.



10.5.6 Wrap-Up: Reflection

Complete each statement based on your current understanding.

1. I am still unsure about Answers will vary.
2. One question I have that will help me better understand is Answers will vary.



For additional data points, consider using student responses from the Activities Stations.

10.6 Comparing Data Sets

Lesson Introduction

 Benchmark	MA.7.DP.1.2 Given two numerical or graphical representations of data, use the measure(s) of center and measure(s) of variability to make comparisons, interpret results, and draw conclusions about the two populations.		 Learning Target	I can compare and interpret two numerical representations of data for two populations.
 Benchmark Coherence	Building On		Working Toward	
	MA.6.DP.1.3 Given a box plot within a real-world context, determine the minimum, the lower quartile, the median, the upper quartile, and the maximum. Use this summary of the data to describe the spread and distribution of the data. MA.6.DP.1.4 Given a histogram or line plot within a real-world context, qualitatively describe and interpret the spread and distribution of the data, including any symmetry, skewness, gaps, clusters, outliers, and the range.		MA.912.DP.2.1 For two or more sets of numerical univariate data, calculate and compare the appropriate measures of center and measures of variability, accounting for possible effects of outliers. Interpret any notable features of the shape of the data distribution.	
 Essential Questions	- How can measures of center and variation be used to compare data sets? - Why do you use different measures of center and variation to describe different data sets? - How do outliers impact the comparison of data sets?			
 Lesson Narrative	This lesson builds on the previous one to continue comparing data sets using measures of center and variation. Students begin by working collaboratively to determine which measures of center and variation should be used to describe data sets and then apply those measures to determine which methods are best suited for various scenarios based on their data presented.			
 Component Summary	10.6.1 Warm-Up (~ 5 min.)	Observe line plots to record what students notice and wonder (<i>SE p. 134</i>)	10.6.4 Activity (10-15 min.)	Apply measures of center and variation to describe and compare two data sets (<i>SE p. 136</i>)
	10.6.2 Exploration (10-15 min.)	Determine measures of center and variation to compare two data sets in context (<i>SE p. 135</i>)	10.6.5 Your Turn! (5-10 min.)	Analyze two data sets to determine the most effective means of communication (<i>SE p. 137</i>)
	10.6.3 Try It (5-10 min.)	Compare data sets to determine the best way to get to school (<i>SE p. 136</i>)	10.6.6 Wrap-Up (~5 min.)	Students explain why they do or do not agree with a conclusion based on the data in the previous component (<i>SE p. 138</i>)
 Resources	Glossary		Materials	
	<u>Key Terms</u> : N/A <u>Additional Terms</u> : N/A		N/A	
			Additional Preparation	
			N/A	

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may struggle to explain their thinking when comparing data sets. - Students may struggle to understand which method of communication is most effective in 10.6.5 Your Turn! activity.	- Consider providing sentence stems for additional scaffolding for written response questions. - Allowing students to use calculators during the lesson will help students focus on comparing and interpreting the data. Consider having no-calculator questions where students turn their calculator over. - If time is an issue, then consider assigning 10.6.5 Your Turn! for homework.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Think-Pair-Share 10.6.3 Try It creates multiple opportunities for students to engage in Think-Pair-Share routines. As students work with their partners, formalize this routine to allow them to formalize their responses before putting them in writing.	MA.K12.MTR.4.1: Discussion and Reflection Partner discussion is embedded throughout this lesson, as almost every component is collaborative. Students are given the opportunity to communicate their mathematical ideas and methodology through discourse, particularly in the 10.6.4 Activity: Guess My Data Sets. Throughout this activity students justify their own reasoning and compare that to the reasoning of their peers.
 Differentiate Instruction	Notice 10.6.4 Activity provides a variety of entry points for learners to engage with the content. This activity is partner-based, offering space for students to verbally reason with a partner and work independently. Some students may prefer to complete problems before discussing with a partner, while others may benefit from verbally processing their reasoning before finalizing their response. All students will benefit from the mathematical discourse.	
Lesson Notes:		

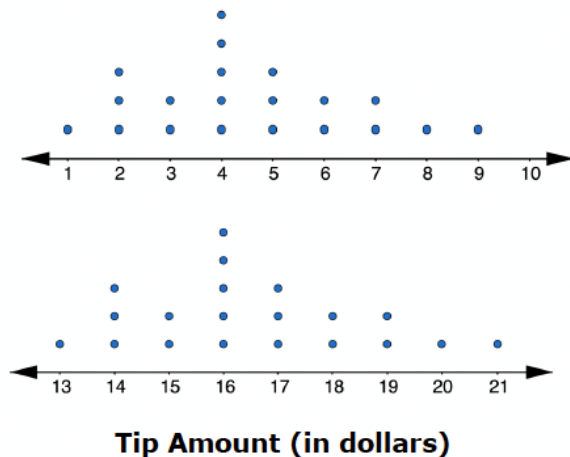
10.6.1 Warm-Up: Notice and Wonder

Component Overview: Students consider two line plots and record what they notice and wonder about what the displays show.



10.6.1 Warm-Up: Notice and Wonder

The line plots represent the distribution of the tip amounts, in dollars, left at two different restaurants on the same night.



Notice and Wonder

Position students to continue to conceptually understand measures of center and variation for two data sets.

Engage students in a Notice and Wonder routine. Ask questions like:

- What do you notice about the data sets?
- What do the line plots have in common?
- What is different?
- What does this make you wonder?

1. Complete each statement.

- a. When looking at the data displays, one thing I notice is ...

Answers will vary. The shape of the distributions is the same, but they are shown over different scales.

- b. One thing I wonder is ...

Answers will vary. Why do workers at one restaurant receive higher tips than the other?

10.6.2 Exploration: Calculating Measures of Center and Variation

Component Overview: Students work with a partner to find the measures of center and variation for two data sets and then analyze the data to draw conclusions about each set. After completing statements describing the data, students explain why the measures of variation and center chosen to summarize the data sets were chosen and check their work with their partner.



10.6.2 Exploration: Calculating Measures of Center and Variation

On a college entrance exam, the possible scores on the math section range from 200-800. Below are the scores for 20 randomly selected female high school seniors.

454 409 437 415 452 440 454 478 401 403 426 426 437
465 465 429 470 448 511 453

The scores for 20 randomly selected male high school seniors are shown below.

408 440 414 391 480 415 426 392 469 446 409 432 375
388 413 451 370 415 487 404

1. Work with your partner to complete the table by calculating the measures of center and variability for each set of data.

Measure	Female Students	Male Students
Mean	443.65	421.25
Median	444	414.5
Range	110	117
IQR	33.5	45
Potential Outliers	511	none

2. Complete the statements.

- a. Female students tend to have

☐ lower
☒ higher

 scores on the math section of the college entrance exam, with a median of 444, than males, who have a median of 414.5.



This Exploration is scaffolded to allow students to engage with it with their partner rather than being guided through by the teacher. There will be an opportunity to summarize and synthesize what is discovered in the next component. Students may revise their thinking throughout the Exploration as they engage with each new Reflection.

Observe and listen as students work with their partners. Make note of student ideas that you can ask them to share in the whole group at the conclusion of the component.

10.6.2 Exploration: Calculating Measures of Center and Variation (continued)

- b. Scores for males tend to have

☐ less
☒ more

 variability,
with an IQR of 45, than scores for females,
with an IQR of 33.5.

- c. Explain why the measures median and IQR were chosen to describe the data sets above, instead of mean and range.
Answers will vary. The mean and range are not accurate measures of center and variation for comparison because 511 is an outlier for the female students' data.
- d. Check your work on Parts A-C with your partner and if necessary, resolve any differences.



Additional scaffolding can be provided through the use of questioning for students struggling to explain:

- *Did outlier(s) exist in the data?*
- *Do outliers play a role in your deciding which measures to use?*

10.6.3 Try It: Drawing Conclusions

Component Overview: Students work with a partner to compare the measures of center and spread for two data sets to determine if a student should ride his bike to school or take a car. Students make arguments for both sides based on the data in the tables.



10.6.3 Try It: Drawing Conclusions

Shawn wants to determine the best way to get to school. Over several weeks, he took turns riding in a car and riding his bicycle and recorded the time, in minutes.

Car	
19	17
15	18
12	22
29	14

Bicycle	
24	25
27	26
21	28
32	23

Shawn calculated the measures of center and spread for each transportation method. They are shown in the table.

	Car	Bicycle
Mean	18.25	25.75
Median	17.5	25.5
Range	17.0	11.0
IQR	6.0	4.0

1. Make an argument as to why Shawn should ride in a car to get to school.

Answers will vary. The average time it takes to get to school by car is less than the average time it takes to travel by bicycle. There is also a larger range in the data, so the times may change, but overall it's faster regardless. Plus, he can leave later in the morning.

2. Make an argument as to why Shawn should ride his bicycle to school.

Answers will vary. Shawn should ride his bicycle to school because the time it takes him to get to school on his bike varies less than the time it takes to travel by car. Even though the car is faster, he can better predict how long it will take to get there on his bike when deciding what time to leave in the morning.



Think-Pair-Share

Engage students in a Think-Pair-Share routine before they complete each written response question. This will give students the opportunity to formalize their responses through discussion before putting them in writing, as well as to receive and provide feedback to partners.



Additional scaffolding can be provided through the use of sentence stems:

- Shawn should take a car/bike to school because
- I know this because the measure of center/variation shows

10.6.4 Activity: Guess My Data Sets

Component Overview: Students work with a partner to play “Guess My Data Sets”. Each partner chooses 2 data sets and writes a paragraph comparing them using measures of center and variability. Both partners read their descriptions while the other partner guesses which two data sets are being described and then records their answers in a table.



10.6.4 Activity: Guess My Data Sets

Suppose an English teacher takes five random samples of students and records their most recent test scores.

Data Set A: {22, 0, 44, 40, 56, 32, 64, 44, 52, 60}

Data Set B: {60, 62, 62, 60, 64, 68, 64, 62, 68, 60}

Data Set C: {70, 82, 74, 76, 80, 72, 84, 70, 78, 82}

Data Set D: {72, 86, 100, 76, 92, 90, 100, 78, 98, 86}

Data Set E: {64, 36, 74, 82, 96, 80, 66, 78, 68, 0}

Without telling your partner, choose two data sets from above.

- Write a paragraph comparing the two data sets you have chosen using measures of center and variability. Make sure to refer to the data sets as “Data Set 1” and “Data Set 2” in the paragraph you write.

Data Set	Mean	Median	Range	IQR	Outliers?
A	41.4	44	64	24	0
B	63	62	8	4	None
C	76.8	77	14	10	None
D	87.8	88	28	20	None
E	64.4	71	96	16	0 or 36

Answers will vary based on data sets chosen. Data Set 1 (C) has a mean of 76.8 where Data Set 2 (D) has a mean of 87.8. The median for Data Set 1 is 77 and for Data Set 2 the median is 88. The IQR for both data sets is 10. Neither data set have any outliers.



Take Turns

Students take turns completing the work in this activity, first explaining the data sets they choose and then choosing which data sets match their partner’s description.

Encourage students to ask any clarifying questions as they work through this component together and justify their descriptions if necessary. As students work, circulate and consider having students summarize their partners’ descriptions.

10.6.4 Activity: Guess My Data Sets (continued)

- Determine with your partner who is partner A and who is partner B. Write your names on the lines for each partner in the table.

Answers will vary.

- Partner A should read their paragraph from Number 1 to partner B, and partner B will guess which two data sets were used. Then, partner B will explain their reasoning for their guess. Both partners will summarize the guess and reasoning in the first row of the table.

Answers will vary.

- Then, switch roles so that partner B reads their paragraph while partner A makes a guess and explains their reasoning.

Answers will vary.

Partner	Guess	Reasoning for Guess
Partner A: _____		
Partner B: _____		



If time allows, challenge students to compare a third pairing of data sets together.

- Reveal to your partner which data sets were used and discuss any errors that may have led to incorrect conclusions.

Answers will vary.

10.6.5 Your Turn!

Component Overview: Students work independently to analyze text and email response times to determine which method of communication is the most effective. If time is an issue, then consider assigning Your Turn! for homework.



10.6.5 Your Turn!

1. A company is trying to determine which method of communication is best to tell their employees important information. They send text messages to nine employees and emails to another nine employees. They record how many minutes it takes for the employees to reply.

Text	5	20	45	9	13	12	4	1	11
Email	25	15	35	24	30	19	29	38	12

- a. Which measure of center best summarizes the data sets?

Text:

- ☐ mean
☒ median

Email:

- ☒ mean
☐ median

- b. Which measure of variability best summarizes the data sets?

Text:

- ☐ range
☒ IQR

Email:

- ☒ range
☐ IQR

- c. Calculate the measures of center and variability.

	Text	Email
Mean	13.3	25.2
Median	11	25
Range	44	26
IQR	12	15.5

- d. Use the most appropriate measures of center and variability to compare the effectiveness of the two methods of communicating with employees.

Answers will vary. Analyzing the data sets measure of center and variability, email appears to be the more effective way to communicate with employees. The mean and median are close together, and there are no outliers in the data set.



Challenge students to make predictions about the efficacy of the two means of communication before finding the measures of center and variability. Have students predict which method will be more effective, then check their work after completing the table.



Students may say that texting is the more effective method of communication because the mean response time is lower than the mean response time for emails, discounting the large variability in text response times. Address this misconception through questioning:

- Is a method more effective if the data is fairly consistent, or if there is more variability? Why?
- Which method is more consistent? How do you know?

10.6.6 Wrap-Up: Cool Down

Component Overview: Students use the data from the previous component to explain whether they agree that email is the best means of communication for the company to use.



10.6.6 Wrap-Up: Cool Down

1. Use the data from Your Turn to complete the prompt below.

Lucas recommends that the company emails its employees because of the smaller range in the data set. Explain why you agree or disagree with Lucas.

Answers will vary. Lucas is correct about the range being smaller.

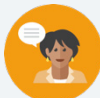


Consider using this Cool Down as a formative data point to inform small groups or partner pairings in the next lesson, as students continue to compare data sets using measures of center and variation.

10.7 Comparing Data Sets and Graphs

Lesson Introduction

 Benchmark	MA.7.DP.1.2 Given two numerical or graphical representations of data, use the measure(s) of center and measure(s) of variability to make comparisons, interpret results, and draw conclusions about the two populations.		 Learning Target	I can compare and interpret a numerical representation of data of one population and a graphical representation of another population.
 Benchmark Coherence	Building On			Working Toward
	MA.6.DP.1.3 Given a box plot within a real-world context, determine the minimum, the lower quartile, the median, the upper quartile, and the maximum. Use this summary of the data to describe the spread and distribution of the data. MA.6.DP.1.4 Given a histogram or line plot within a real-world context, qualitatively describe and interpret the spread and distribution of the data, including any symmetry, skewness, gaps, clusters, outliers, and the range.			MA.912.DP.2.1 For two or more sets of numerical univariate data, calculate and compare the appropriate measures of center and measures of variability, accounting for possible effects of outliers. Interpret any notable features of the shape of the data distribution.
 Essential Questions	- Is it easier to compare data presented graphically or in numerical data sets? Why? - How can you compare data sets without finding all the measures of center and variation? - How can you use measures of center and variation to draw conclusions about different data sets?			
 Lesson Narrative	In this lesson, students compare data presented numerically and graphically. Students begin by summarizing data presented numerically and in a box plot, considering the benefits and drawbacks of presenting data numerically and graphically. Students then participate in a dice-rolling activity to collect their own data and compare it to a partner's. The lesson culminates with students summarizing two data sets to compare them and draw conclusions.			
 Component Summary	10.7.1 Warm-Up (~5 min.)	Students choose between two deals (<i>SE p. 139</i>)	10.7.4 Activity (10-13 min.)	Work in pairs to roll dice, record the data, and compare the results (<i>SE p. 142</i>)
	10.7.2 Guided Practice (10 min.)	Summarize data presented in different ways (<i>SE p. 140</i>)	10.7.5 Your Turn! (10-12 min.)	Compare and analyze two different data sets (<i>SE p. 143</i>)
	10.7.3 Try It (5-10 min.)	Compare data presented in different ways (<i>SE p. 141</i>)	10.7.6 Wrap-Up (~5 min.)	Explain how to compare a data set to a graphical representation of data (<i>SE p. 145</i>)
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> N/A		- Dice (one for each pair of students) - If dice are not available, have students use random.org and select one die as the option	(Optional) Consider choosing student pairs ahead of time based on data from prior lessons

 Teacher Tips	Common Misconceptions - Students may struggle to compare data sets without first calculating the measures of center and variation. - Students may struggle to explain their answers in writing.	Things to Consider - Consider providing writing prompts or sentence starters for questions that require written explanations. - If time is an issue, then consider assigning all or part of the 10.7.5 Your Turn! for homework.
	Literacy and Language Routines Think-Pair-Share 10.7.2 Guided Practice provides an opportunity to engage the class in a Think-Pair-Share routine. This will give students time to contemplate why comparing data graphically may be easier, while also considering the drawbacks of box plots in terms of information that can be collected. As students share with a partner, circulate and encourage students to build on to or evaluate their partner’s response.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.7.1: Real-World Contexts Throughout this lesson, students work collaboratively and independently to analyze data from real-world contexts. This encourages students to conceptually understand measures of center and variation and how they can be applied to make sense of data in the real world. Clarifications for this standard include providing students with opportunities to create models and perform investigations, which students are given the opportunity to do through the 10.7.4 Activity.
	 Differentiate Instruction Throughout this lesson students are provided with multiple entry points to engage with the content. Visual displays of data are presented throughout to help students make sense of the data sets. Written explanations are intentionally scaffolded through the use of word banks, and this will be particularly helpful for ELL students to reinforce the meaning of the terms through the connection to visual data displays. Partner work is built in throughout this lesson, providing auditory entry points and means for verbal processing of information through discussion.	
Lesson Notes:		

10.7.1 Warm-Up: Would You Rather?

Component Overview: Two deals are presented, and students explain which deal they would rather take.



10.7.1 Warm-Up: Would You Rather?

1. Would you rather...

- A. Pay for 3 gifts and get 1 of equal or lesser value free?
- B. Pay for 4 gifts using a coupon for 20% off?

Explain your reasoning.

Answers will vary. If the gifts cost \$5 each, a person selecting option A would pay \$15 and a person selecting option B would pay \$16. Option A would be the better choice.



Challenge students who finish early to prove why their choice is the right choice, using examples, and consider whether there are circumstances that would change which deal they would take.

10.7.2 Guided Practice: Summarizing Data

Component Overview: Students compare travel times for two counties given data sets represented in different ways.

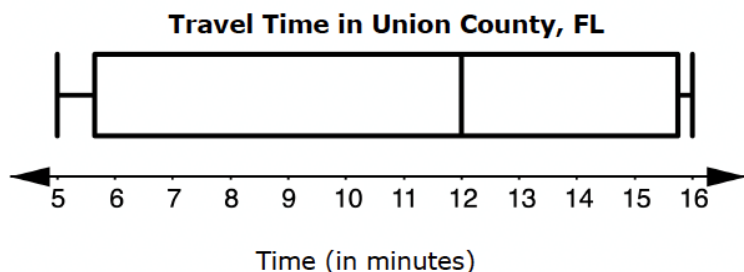


10.7.2 Guided Practice: Summarizing Data

- The commute times, in minutes, for 24 randomly selected people in Duval County, FL is shown.

9 10 15 17 18 20 20 21 22 23 24 25
25 25 25 26 26 29 29 29 32 34 38 40

The box plot displays the commute time, in minutes, for 24 randomly selected people in Union County, FL.



Think-Pair-Share

Pair students and engage the class in a Think-Pair-Share routine as they answer questions 1a and 1b. This will give students time to contemplate why comparing data graphically may be easier, while also considering the drawbacks of box plots in terms of information that can be collected. As students share with a partner, circulate and encourage students to build on to or evaluate their partner's response.

- Explain why it would be beneficial to create a box plot of the Duval County data when comparing these data sets.

Answers will vary. Having a box plot of Duval County would make comparing the data easier because both data sets will be in the same form.

- What are some measures used to describe data that cannot be determined from a box plot?

Answers will vary. Mean and mode

10.7.2 Guided Practice: Summarizing Data (continued)

- c. Use at least three terms from the term bank below to write a three-sentence summary comparing travel times to work in Duval County to travel times to work in Union County.

mean	median	range	IQR
center	variability	upper quartile	
lower quartile	sample	population	

Answers will vary. The median in Duval County is 25 minutes where the median in Union County is 12 minutes. It takes longer to travel to work in Duval County than it does in Union County. The IQR for Duval County is $29 - 20 = 9$ minutes, and the IQR for Union County is about 10 minutes, based on the box plot. The variability of the data can be compared through the range of each data set. The travel times vary less in Union County because the range is 11 minutes whereas the range in Duval County is 31 minutes.



Question 1c is scaffolded through the use of a word bank to provide students with key terms to consider as they explain their response. If you notice students struggling with their answer, further scaffold with questions like:

- Which measures of center can we find for each data set? What will this tell us?
- Which measures of variability can we find for each data set? What can this tell us?
- What can you say for sure about the travel times of each county? Why?

10.7.3 Try It: Comparing Data Sets and Graphs

Component Overview: Students work independently or with a partner to compare two data sets presented in different formats.



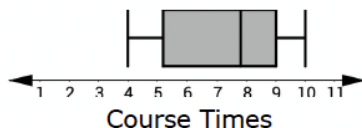
10.7.3 Try It: Comparing Data Sets and Graphs

- The Deltona High School Wolves and Wildwood High School Wildcats cross-country teams ran an obstacle course. Each team's times are summarized.

Deltona HS Wolves' Obstacle Course Times

4: 25	4: 43	4: 49	5: 02	5: 12
5: 21	5: 31	5: 32	5: 37	5: 52
5: 54	6: 08	6: 20	6: 26	6: 33
6: 48	6: 53	7: 16	7: 23	8: 05

Wildwood HS Wildcats' Obstacle Course Times



- Which statements are true about the data for the Deltona Wolves and the Wildwood Wildcats? Select all that apply.
 - ☐ The median time of the Deltona Wolves is less than the median time of the Wildwood Wildcats.
 - ☐ The fastest 25% of athletes on both teams complete the obstacle course in about the same amount of time.
 - ☐ The interquartile range of the Deltona Wolves is less than the interquartile range of the Wildwood Wildcats.
 - ☐ Approximately 50% of Wildwood Wildcats have times between 5 and 6 minutes.
- Which team do you think is more likely to win the next race? Explain your reasoning.
Answers will vary. Based on the data, Deltona will win the next race. Their course times vary less than Wildwood and all fall under Wildwood's 10 minute mark. Analyzing each team's IQR, Deltona's spread is smaller.



Students may struggle to compare the data sets, particularly when reasoning about the measures of center and variation to choose the true statements. Use strategic questioning to scaffold this section for students:

- How can you determine the median of the data sets?
- Where are the fastest 25% of athletes located in each data set?
- How can you determine the IQR for both teams?



It may be helpful for students to visualize Deltona's times as "quartiles," similarly to how data is presented in the box plot. The data is organized into 4 rows increasing in times from left to right. Encourage students to circle or label each row. The first row corresponds to the left whisker of a box plot, the second row corresponds to the first quartile, the third row corresponds to the third quartile, etc. Though this is not exact, the visual and labels can be a starting point for students as they reason about which statements are true in question 1a.

10.7.4 Activity: Roll the Dice

Component Overview: Students work in pairs to play “Roll the Dice.” Each partner rolls the die 25 times, and both partners record the results either on a line plot or numerical data set before comparing their data sets.



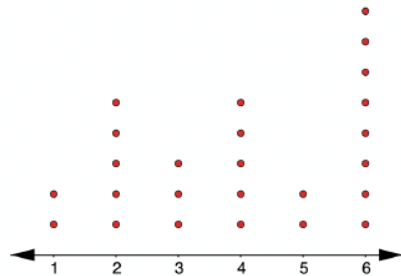
10.7.4 Activity: Roll the Dice

With your partner, decide which of you will be “Partner 1” and “Partner 2.” Partner 1 will be creating a line plot and Partner 2 will be recording a data set.

Using a standard six-sided die, each person will roll one die 25 times. After each roll, record the results in either a line plot or numerical data set below, depending on if you are Partner 1 or Partner 2.

- When both partners are finished, copy their data below.
Answers will vary.

Line plot (Partner 1’s rolls): *Answers will vary.*



Data set (Partner 2’s rolls): *Answers will vary.*

1	2	6	3	6
6	2	4	6	6
6	1	3	5	6
3	5	5	5	5
1	2	5	4	4



If dice are not available, have students use random.org and select one die as the option.



This activity provides auditory, visual, and kinesthetic entry points. Learners have a variety of ways to engage with the content and expand their understanding of how to compare data sets.

10.7.4 Activity: Roll the Dice (continued)

2. Complete the table by calculating the measures of center and spread for your data. Share your data with your partner and add their data to the table.

Answers will vary.

	Partner 1	Partner 2
Mean	3.96	4.08
Median	4	5
Range	5	5
IQR	4	3.5

3. With your partner, discuss similarities and differences in your data. Summarize your discussion in the table.

Answers will vary.

Similarities	Differences
<i>The range for both is 5. Both partners rolled 25 times. The means are both close to 4.</i>	<i>Partner 2 had a larger median but a smaller IQR.</i>



Consider providing guiding questions for students to reference as they discuss the similarities and differences between their data sets. This will encourage participation and help guide thinking towards deeper understanding.

- How do the means of the data sets compare? What does that tell you?
- What are the medians of each data set? What does that tell you?
- Which measures of center/variation best summarizes each data set? Why?

10.7.5 Your Turn!

Component Overview: Students work independently to compare data sets represented in different ways.



10.7.5 Your Turn!

1. A group of ten rowers that practice outdoors on the water (Group 1) are challenging a group of ten who use indoor rowing machines (Group 2). They want to see who can row faster using rowing machines.

The time, in minutes, it took each of the ten rowers in Group 1 to row 500 meters on the rowing machine is represented in the stem-and-leaf plot.

Group 1: Outdoor Rower Times (in minutes)

Stem	Leaf	
15	2, 8	<i>Mean: 1.733 minutes</i>
16	8	<i>Median: 1.765 minutes</i>
17	0, 5, 8	<i>Range: 0.35 minutes</i>
18	0, 2, 3, 7	<i>IQR: 0.14 minutes</i>

Key: 15|2 = 1.52



After students have read the context, consider prompting them to determine the measures of center and variation for each data set to prepare them for the questions that follow.

The time, in minutes, it took each of the ten rowers in Group 2 to row 500 meters on the rowing machine is represented in the data set.

{1.63, 1.82, 1.95, 1.97, 2.0, 2.02, 2.12, 2.20, 2.25, 2.88}

Mean: 2.084 minutes
Median: 2.01 minutes
Range: 1.25 minutes
IQR: 0.25 minutes

10.7.5 Your Turn! (continued)

For each statement below, state whether it is true or false. For any statements that are true, explain why.

Statement	True or False?	Explanation
The center of the Group 1 times is similar to the center of the Group 2 times.	<i>False</i>	<i>The center for Group 1 is lower than Group 2. Both the mean and median for group 1 is smaller than the mean and median, respectively, for Group 2.</i>
The variability of the Group 1 times is similar to the variability of the Group 2 times.	<i>False</i>	<i>The variability is different for the groups, regardless if the range or IQR is used to compare. Both the range and IQR are smaller for Group 1 compared to the same measure for Group 2.</i>
The Group 1 rowers that practice on the water are typically faster than the Group 2 rowers that row on machines.	<i>True</i>	<i>The rowers that practice on water, Group 1, have faster times than those who row indoors, Group 2. The measures of center for Group 1 are smaller than Group 2, indicating a faster time.</i>
The Group 1 rowers that practice on the water are more consistent than the Group 2 rowers that row on machines.	<i>True</i>	<i>Answers will vary. Both measures of variability for Group 1 are low, indicating their times are not as spread out from one another.</i>



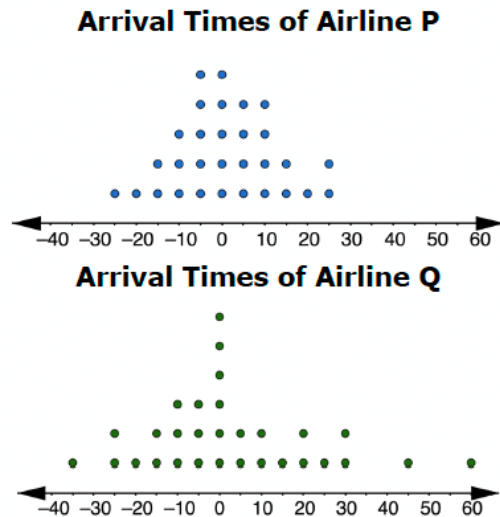
Students do not need to write explanations for the false statements. Sample responses are provided for reference, though.



The students are not directly told to calculate center and variability. Instead, they are asked questions about the summary of the data sets. If students are struggling, it may be helpful to have them determine the measures of center and variability for each data set.

10.7.5 Your Turn! (continued)

2. The dot plots display the arrival times of 30 flights for two different airlines.



- A negative number represents the number of minutes the flight arrived before its scheduled time.
- A positive number represents the number of minutes the flight arrived after its scheduled time.
- A zero indicates that the flight arrived at its scheduled time.

Based on the data, from which airline would you choose to buy your ticket? Use measure(s) of center and variability to justify your choice.

Answers will vary.

Airline P – mean: 1, median: 0, range: 50, IQR: 15

Airline Q – mean: 3.17, median: 0, range: 95, IQR: 25

Airline P has less variation with an IQR of 15 compared to Airline Q which has an IQR of 25. Airline P also has an average arrival time of 1 minute after their scheduled time where Airline Q has an average arrival time over 3 minutes after their scheduled arrival time. Based on the data, Airline P would be the better choice to buy a ticket from.



Students may struggle to understand what negative numbers and 0 mean in the context of the flight times and therefore choose the airline with the higher measures of center. If you notice students making this mistake, ask questions to foster contextual understanding:

- *Have you ever flown on a plane? Did you arrive on time?*
- *Is it better to have a flight time longer or shorter than expected? Why? Which numbers indicate longer flight times? Shorter?*
- *If you want to arrive on time or before schedule, which airline should you choose based on the data? Why?*

10.7.6 Wrap-Up: Cool Down

Component Overview: Students summarize their experience from the lesson.



10.7.6 Wrap-Up: Cool Down

1. Write a short note to a classmate who missed class summarizing today's lesson.

Answers will vary.

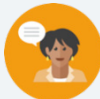


Consider pairing students to discuss their notes. Use student responses as anchor charts or added to scaffolding notes.

10.8 Random Samples vs. Population

Lesson Introduction

 Benchmark	MA.7.DP.1.3 Given categorical data from a random sample, use proportional relationships to make predictions about a population.		 Learning Target	I can determine if a sample is representative of the population.
 Benchmark Coherence	Building On N/A		Working Toward MA.912.DP.1.4 Estimate a population total, mean, or percentage using data from a sample survey and develop a margin of error through the use of simulation.	
 Essential Questions	- How can you collect data about a large population? - How do you choose a sample? - How do you ensure a sample is representative of the population?			
 Lesson Narrative	In this lesson, students are introduced to surveying a population using a representative sample. They are guided through understanding the difference between the population and a sample and determining the best means for ensuring that they are collecting data from a representative sample. Students then collaborate and explore to further develop the idea of representative samples, including exploring how the size of the sample can impact its accuracy reflecting the population.			
 Component Summary	10.8.1 Warm-Up (~5 min.)	Career Profile – electric vehicle technician (<i>SE p. 146</i>)	10.8.4 Exploration (10 min.)	Explore how sample size affects results (<i>SE p. 149</i>)
	10.8.2 Guided Instruction (10 min.)	Introduction to choosing representative samples of a population (<i>SE p. 146</i>)	10.8.5 Your Turn! (5 min.)	Choose, describe, or critique sampling methods used for given populations of interest (<i>SE p. 150</i>)
	10.8.3 Collaboration (10-15 min.)	Collaborative exercise in comparing sampling methods and suggesting better methods (<i>SE p. 148</i>)	10.8.6 Wrap-Up (~5 min.)	Student assessment of choosing a representative sample (<i>SE p. 151</i>)
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> - Population - Sample - Representative - Random sampling <u>Additional Glossary:</u> N/A		- Career Profile video (Optional) Calculator (Optional) Premade circles for 10.8.4 Exploration (Optional) Markers	10.8.1 Warm-Up contains a video to accompany the question prompts.

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may confuse population with sample. - Students may struggle with the idea of a sample being representative of a population. - Students may not be able to predict how different methods of sampling could lead to different and misleading conclusions.	- Once students have worked on the first problem polling voters in Florida, consider stopping and doing a check with them to make sure they are on the right track. - Consider creating a visual representation of the new vocabulary to use as an anchor chart or on a word wall.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Think-Pair-Share 10.8.3 Collaboration gives students the perfect opportunity to apply these new concepts to different contexts. Consider applying the Think-Pair-Share routine for both scenarios to create opportunities for students to solidify their understanding of the difference between a sample and a representative sample.	MA.K12.MTR.4.1: Discussion and Reflection Students are being introduced to the idea of samples of a population as well as representative samples. Because this lesson is based around the vocabulary, guide students to discuss and reflect on these ideas in order to more fully develop them. Both the 10.8.3 Collaboration and 10.8.4 Exploration provide opportunities for students to discuss and reflect on their ideas about these concepts in context.
 Differentiate Instruction	10.8.4 Exploration implements both auditory and visual methods into the activity. Consider providing drawing implements for students to complete their spinners to aid in their understanding. Additional differentiation opportunities are denoted throughout the lesson guide.	
Lesson Notes:		

10.8.1 Warm-Up: Electric Vehicle Technician

Component Overview: Students watch a career video about Ben Grandy, an electrical vehicle technician with a background in landscape architecture and electrical engineering. After the video “Day at Work: Electric Vehicle Technician” students reflect on some of Grandy’s values concerning careers and life in general.



10.8.1 Warm-Up: Electric Vehicle Technician

Ben Grandy is an electrical vehicle technician for City Car Share. When Ben was considering what to do with his life, he knew he wanted to do something that would bring him happiness. Now he uses the skills he has gained from his study of landscape architecture and work as an electrical engineer to maintain the fleet of electric vehicles and electric bikes used by City Car Share.

Ben found that dedicating himself to a cause that is bigger than himself brings him happiness.

1. Think about what you would like to do with your life. What kind of work would make you happy?
Answers will vary. I would like to be a teacher because I like to help people. I would like to be a doctor because I’m curious about the brain.

In talking about his work, Ben said, “That’s how engineering and design is. You never get to a point where you think, ‘Oh, that’s perfect! That’s done.’ You start asking questions like, ‘Now that we’ve gotten this far, what’s the next step? How can we improve this?’”

2. How do you think having this mentality is helpful?
Answers will vary. This mentality is helpful because it means you can always get better. It is helpful because it means that you will never be lazy or complacent.



Do you know someone who is an electrical engineer or an electric vehicle technician? Consider bringing in a visitor for your class to ask questions and learn from. Additionally, consider allowing students to research fields where electrical engineers can work, choosing one of interest and creating a summary based on their learning either for homework or extra credit.

10.8.2 Guided Instruction: Is the Sample Representative of the Population?

Component Overview: The difference between populations versus a sample is introduced to students and cemented by having students determine the population and sample for various scenarios.



10.8.2 Guided Instruction: Is the Sample Representative of the Population?

Many organizations want to know how large groups of people think and behave. This helps businesses make decisions regarding what to sell, when to be open, and whom to target with advertisements. Research is also used to make predictions about events like elections, and to help doctors provide appropriate healthcare to their patients.

To gather this information, companies and organizations often collect data **samples** from a **population** they are studying.



A **population** is the entire set of cases or individuals under consideration in a statistical analysis.

A **sample** is a subset, or part, of a population.

1. Imagine a group of researchers want to gather information about which candidate for sheriff all registered voters in Glades County, Florida plan to vote for in an upcoming election.

Use the definitions to complete the statements.

In this scenario, the ☐ population ☐ sample being studied is

all registered voters in Glades County, FL. It is not realistic for researchers to ask every single registered voter, so researchers would collect data from a

☐ population ☐ sample

of registered voters in Glades County, FL to make predictions.



This is the first time that students are introduced to the terms population, sample, representative sample, and random sampling. While modeling the problems in this component, include developing a strong understanding of these terms.



- What is the group the researchers are studying? Does this group represent the population or the sample?
- Whom are the data from? Does this group of people

10.8.2 Guided Instruction: Is the Sample Representative of the Population? (continued)

2. Volusia County School District wants to find out more about which mode of transportation students take to school, so they decide to conduct a survey. The entire school district has a population of 63,223 students. Since the number of students is so large, the district will only survey some of the students.

- a. In this scenario, what is the population being studied?

Students in the Volusia County School District

- b. What is the sample being used?

The students that are interviewed for the study.

To ensure the students surveyed are representative of all students in Volusia County, a system is used to select who is surveyed.



A sample that is **representative** of a population resembles the population and can be used to make accurate predictions about the population.

3. Twenty-five students from each grade level will be chosen at random to participate in the survey. With your partner, discuss how you think this method will help ensure the sample is representative of the population.

Answers will vary. This will help ensure that the sample is not biased, however that size is a bit small to ensure that there is true variability.



Consider having students connect the idea of a representative sample to the idea of representatives in real life. The whole school doesn't meet to make decisions; instead, a representative of each grade is elected to a council to speak for that grade. Each state can't send its entire population to the house of representatives to vote, but the representative goes and votes for them.



A representative sample is one where all parts of the population are represented.

- *Would it be a good representation if only the boys were surveyed?*
- *Or what about only the people with fall birthdays?*
- *How can we determine what a good representative sample might be?*

10.8.2 Guided Instruction: Is the Sample Representative of the Population? (continued)

4. Claire wants to collect data on the highest education level achieved by adults in Jacksonville, FL. She decides to go to the University of North Florida campus and ask the first person who comes into the library every 30 minutes about their education level.

- a. What is the population?

The population is adults in Jacksonville, Florida.

- b. What is the sample?

The sample is the first person coming into the UNF library every thirty minutes.

- c. Explain why a sample chosen in this way may **not** be representative of the population.

The sample will most likely be college-aged students, rather than adults of all ages. There also may be a disproportionate representation of college-educated adults since it is taking place on a college campus.



If students struggle to explain why the sample is not representative, guide their thinking through questioning:

- *If Claire is asking about education levels on a college campus, who will she most likely be talking to?*
- *If someone already has a degree, why would they be in the campus library?*
- *If someone chose a line of work that does not require a degree, why would they be in a campus library?*



Consider challenging students to discuss how Claire could find a representative sample to survey.



A **random sampling** is a smaller group of people or objects chosen from a larger group or population by a process giving equal chance of selection to all possible people or objects, and all possible subsets of the same size.

10.8.3 Collaboration: Comparing Methods for Selecting Samples

Component Overview: Students collaborate to analyze different survey methods. Using their analyses, students discuss how their survey methods could impact the conclusions drawn. They then work together to recommend how to find a sample that is more representative of the population.



10.8.3 Collaboration: Comparing Methods for Selecting Samples

For each scenario below, read the information about the study and answer the questions that follow.

Scenario 1: Lin is running for president of the seventh grade. She wants to predict her chances of winning the election. She is considering three different methods for surveying a sample of the students.

1. Complete the table to describe the benefits and problems with each of the possible survey methods described.

Sample Surveyed	Benefits of this Method	Problems with this Method
Method 1: Ask everyone on her basketball team who they are voting for.	<i>This method is convenient.</i>	<i>Her teammates may not respond to her honestly. There may be students who aren't in seventh grade on her team.</i>
Method 2: Ask every third girl waiting in the lunch line who they are voting for.	<i>This will provide a good sample size.</i>	<i>The sample is not representative of the population – only female, and they may not be seventh graders depending upon the lunch period.</i>
Method 3: Ask the first 15 seventh graders to arrive at school one morning who they are voting for.	<i>This method includes the voting demographic (seventh graders).</i>	<i>Students who arrive early at school may be part of the same subgroup (arriving early for a club meeting or practice). It is also a small sample size.</i>



Think-Pair-Share

To encourage student discussion, consider having students read each scenario first and use a Think-Pair-Share to discuss the answers.

Students should be collaborating throughout the activity. Monitor to identify any misconceptions, especially in regard to representative samples, to identify any students who might need further clarification.

10.8.3 Collaboration: Comparing Methods for Selecting Samples (continued)

2. How could the different methods for selecting a sample in Scenario 1 lead to different conclusions about the population?

Method one would highlight how basketball players feel about the election – they likely would vote for Lin because they are teammates, so this would be a biased sample. Also, not all of these students would be able to vote for her as they may be in different grades.

Method two would provide a sample of girls of different grades at lunch time. It does not include boys, who may vote differently, and it may include some non-seventh graders.

Method three selects the target grade group, but might only include students who are dropped off / walk to school. It is also a very small sample, so the feelings of the general seventh grade population may be very different.

3. Explain which of the methods listed in the table would be the most likely to produce a sample that is representative of the population.

The third method is the most targeted – it specifies seventh graders. It could be made stronger if Lin chose students at random throughout the morning – students arriving at the same time are likely to be neighbors or friends.

4. With your partner, determine a better way to select a sample for this study.

Student responses will vary – a better way to select a sample would be to target random groups of seventh graders – perhaps to survey every fifth seventh grader that enters in the morning.



For students who may still be struggling with the idea of population versus representative sample, consider asking:

- Who or what is the population?
- Who or what would the sample be?
- Who or what does the sample NOT include?

By asking who or what is left out of the sample, students can think of “representative” in another way, for example, as in not leaving specific groups of the population out.

10.8.3 Collaboration: Comparing Methods for Selecting Samples (continued)

Scenario 2: Maneet, a nutritionist, wants to collect data on how much caffeine the average American drinks per day. She works with three of her colleagues to decide how to survey a sample of the population.

Method 1	Method 2
Maneet and Diarmaid believe the best way to survey is to ask the first 20 adults who arrive at a grocery store after 10:00 A.M. about the average amount of caffeine they consume each day.	Faith and Santiago would rather ask the first adult who comes into a coffee shop every 30 minutes about the average amount of caffeine they consume each day.



For students who finish early, consider directing them to find another way to choose a representative sample for scenario 2.

5. What recommendation(s) would you give to this committee about how to select the most representative sample?

In the first method, you are likely to have a sample that has similar profiles (early morning coffee drinkers.) While it's better to conduct the survey at a neutral location like a grocery store, it would be better to conduct it throughout the day and with more random selections (i.e. every ___th adult who comes in).

With the second method, there is likely going to be some bias because the survey is being conducted at a coffee shop. Adults going to a coffee shop may be likely to consume more caffeine. They should change their location but keep the randomness of their selection method.

10.8.4 Exploration: Sample Size

Component Overview: Students explore how sample size can affect understanding of the population as a whole. Encourage students to discuss how they could determine what the spinner would look like. Encourage students to compare the first and the second sample data. Emphasize that if a sample is too small, then important details about the population may be missed.



10.8.4 Exploration: Sample Size

Isabella is playing a computer game. Her challenge is to create a replica of a hidden spinner that she cannot see. Her score is based on two things: the accuracy of the spinner she creates and the number of spins she uses to create the spinner (the fewer spins she uses equals a higher score).

Isabella creates her first replica after 12 spins of the hidden spinner. Below is a table showing the outcomes of the first 12 spins.

Color	Number of Spins
red	2
purple	4
blue	3
yellow	3

- Use the data from the first 12 spins to draw a sketch of what you think the hidden spinner might look like.



The objective is not to have a perfect circle but rather a model that represents the data in the table. It may be helpful to have students identify the population (the spinner) and the sample (the 12 spins) before completing the problem in order to connect the lesson's concepts.



As stated above, the objective isn't about the circle so much as the model. Therefore, it may be helpful for students that are struggling to be provided with an already created circle that is split into 12 sections in order for students to focus more on the idea of the sample versus interpreting data.

10.8.4 Exploration: Sample Size (continued)

Isabella's spinner was not accurate, so she had the computer generate additional data. The table below shows the data from the first 36 spins of the spinner, including the 12 shown in the previous table.

Color	Number of Spins
red	3
purple	10
blue	8
yellow	4
orange	2
green	9

2. Revise your spinner based on this new data.



3. Explain which spinner you think is closest to what the hidden spinner looks like.

The second spinner seems more accurate. There were more colors than we had previously known, and the distribution was more varied. There was also more data available to construct the second spinner. A larger sample gives more information.

4. The second spinner Isabella drew was far more accurate than the first. Having additional spins gave a more accurate representation of what the spinner actually looked like. What can this simulation teach us about choosing the size of the sample in a population study?

This shows us that it's important to have a large sample – otherwise we can draw inaccurate conclusions.



Students don't need to worry about perfectly splitting the circle into 36 sections. If students are struggling, facilitate thinking of the number of spins as part of a whole. Therefore, the green spins were 9 out of 36 which is equivalent to a fourth of the circle. Choose the data that is easiest to visualize to section the circle first, then fill in with the other sections. Again, consider providing a circle to students if desired.



Facilitate student discussion toward the understanding that too small of a sample can lead to part of the population not being represented at all.

- How does this table compare with the last?
- Which colors weren't even represented in the first table?
- Looking at the data, which color would you assume takes up the most space on the spinner? And the least space?
- How does that compare with your first spinner?

10.8.5 Your Turn!

Component Overview: Students recommend the best way to create or provide feedback on samples so that they are more representative of the population in multiple situations. Consider allowing students to work with partners in order to allow verbalization of ideas before presenting a final answer.



10.8.5 Your Turn!

1. The meat department manager at a grocery store is worried some of the packages of ground beef labeled as having one pound of meat may be under-filled. He decides to take a sample of 5 packages from a shipment containing 100 packages of ground beef. The packages were numbered as they were put in the box, so each one has a different number between 1 and 100.

Describe how the manager can select a fair sample of 5 packages.

The manager could choose multiples of 20. Then he would have five samples that had been spaced out throughout the order.

2. Diego wants to survey a sample of students at his school to learn about the percentage of students who are satisfied with the food in the cafeteria. He decides to go to the cafeteria on a Monday and ask the first 25 students who purchase a lunch at the cafeteria if they are satisfied with the food. Explain whether you think this is a good way for Diego to select his sample.

Diego should consider choosing students at random rather than the first twenty-five students. The students who arrive early to lunch are most likely the ones who are excited about lunch and would skew his data positively.

3. Eleanor wants to make the most accurate prediction about who will win the student council election. Explain whether it would be better for her to conduct a random sample of 10 people or 50 people.

A sample of fifty people would provide a bigger and thus more accurate predictive sample.



For students who might struggle verbalizing the best way to choose a representative sample, provide sentence stems or targeted questions for each scenario.

- How could he ensure that he is checking packages randomly throughout the order?
- The meat department manager could choose to sample every _____ package.
- What kind of opinions might the first 25 people in line have?
- If Diego chooses the first 25 students in line, _____.
- How does the size of a sample impact how well it represents the population?
- Eleanor should choose _____ because it is likely to better represent the population as it is a _____ sample.

10.8.6 Wrap-Up: Cool Down

Component Overview: Students will complete a Cool Down demonstrating the student's understanding of choosing an appropriate sample. This activity will provide instructional data and therefore should be completed independently.



10.8.6 Wrap-Up: Cool Down

Anthony is conducting research to determine how students at his school get to school each day (biking, walking, car, bus, etc.). Because there are many students in his school, he will be conducting a survey of a random sample of students to make predictions about the overall population.

1. Select all the options that would produce a random sample to represent the overall population of the school.
 - ☐ Anthony will get to school before everyone else and survey the first 50 people who walk in the doors.
 - ☐ Anthony will survey everyone at his jazz band practice.
 - ☐ Anthony will survey two randomly chosen students in every homeroom.
 - ☐ Anthony will assign each student in the school a number, randomly choose 50 numbers, and survey those students.
 - ☐ Anthony will survey every third student he sees on his walk to school.

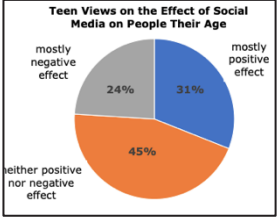


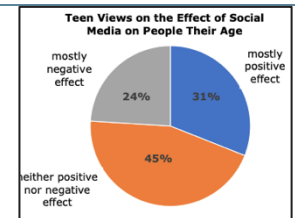
For students who may finish early, challenge them to describe how the other options could result in misleading survey results.

10.9 Making Population Predictions - Part 1

Lesson Introduction

Lesson Introduction						
 Benchmark	MA.7.DP.1.3 Given categorical data from a random sample, use proportional relationships to make predictions about a population.			 Learning Target	I can use proportional relationships to make predictions about a population.	
 Benchmark Coherence	Building On			Working Toward		
	N/A			MA.912.DP.1.4 Estimate a population total, mean, or percentage using data from a sample survey and develop a margin of error through the use of simulation.		
 Essential Questions	- How can data from a sample be used to make predictions about a population? - How can fractions and percentages be used to make population predictions?					
 Lesson Narrative	Given data in multiple forms and using some new situations, students use the data to calculate percentages of the population that fit the given situations. They are introduced to three different strategies to use for making predictions about the populations using calculated percentages from sample data. The lesson provides opportunities for them to understand and analyze each strategy as they determine which they prefer or which works best for different situations.					
 Component Summary	10.9.1 Warm-Up (~5 min.)	Notice and Wonder about a restaurant's prices (<i>SE p. 152</i>)		10.9.3 Exploration (10 min.)	Explore how to use data from a random sample to make predictions (<i>SE p. 154</i>)	
	10.9.2 Guided Instruction (10-15 min.)	Learn to use equivalent fractions, scale factor, and percentages to make predictions about a population from a sample (<i>SE p. 152</i>)		10.9.4 Guided Practice (10-15 min.)	Synthesize learning to practice problems with teacher's support (<i>SE p. 155</i>)	
	10.9.5 Wrap-Up (~5 min.)	Reflect on learning and understanding of today's learning target (<i>SE p. 157</i>)				
 Resources	Glossary		Materials		Additional Preparation	
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> N/A		N/A		N/A	

 Teacher Tips	Common Misconceptions - Students may get confused because different strategies for making population predictions may result in slightly different answers. - Students may also get confused by the way the word “estimate” is used in this lesson. They need to understand that the prediction is an estimate because it can’t be known for certain, but they should still be finding exact values based on the predicted percentages from the sample data.	Things to Consider - At this time, the focus in problem-solving for these types of questions is NOT on using proportions. Unit 12 will focus on setting up and solving proportions. The focus in this unit is on using strategies such as equivalent fractions, scale factor, and percentages. - In problems that require several steps, students should not round until the final step. Rounding midstep could significantly change the answer. If not given a place value for rounding, then determine what place value you would like your students to round to. Many assessments have students round to three decimal values. Incorporating the Desmos scientific calculator can add a powerful tool for students who struggle with completing steps on a handheld calculator. The Desmos calculator allows students to input complicated arithmetic that would require several steps on a handheld calculator. This eliminates common, careless errors that many students make.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Notice and Wonder The 10.9.1 Warm-Up allows students to explore, wonder, and discuss their mathematical ideas in regards to a restaurant menu. It can open up the idea of analyzing differences and discovering if changes are consistent (proportional).	MA.K12.MTR.6.1: Reasonableness 10.9.3 Exploration gives students an opportunity to make predictions and reflect on the reasonableness of their predictions as well as those made by other students using different strategies.
	Differentiate Instruction The 10.9.3 Exploration allows the teacher the opportunity to make informal observations to inform groupings during 10.9.4 Guided Practice. Consider whether students are ready to work independently or with partners and which partnerships would best support all students. It may also be appropriate to pull a small group during the 10.9.4 Guided Practice based on observations during the 10.9.3 Exploration. Additional differentiation opportunities are denoted throughout the lesson guide.	Teen Views on the Effect of Social Media on People Their Age 
Lesson Notes:		



10.9.1 Warm-Up: Notice and Wonder

Component Overview: Given a menu, students observe and analyze price differences through a Notice and Wonder routine.



10.9.1 Warm-Up: Notice and Wonder

Burgers	Chips	Fries
Deluxe Burger	\$6.90	\$7.70
Deluxe Cheese	\$7.70	\$8.45
Bacon Burger	\$7.95	\$8.65
Bacon Cheese	\$8.25	\$8.95
Double Burger	\$8.35	\$9.30
Double Cheese	\$9.25	\$9.95
Double Bacon Cheese	\$9.50	\$10.25
Cheese Garden Burger	\$8.25	\$8.85
Mushroom Swiss	\$8.25	\$8.95
Jalapeno Pepper Jack	\$8.25	\$8.95
BBQ Western Bacon	\$9.20	\$9.85
Pastrami Burger	\$9.20	\$9.85
Power Pac	\$10.45	\$11.05
Cheese Power Pac	\$11.25	\$11.70

1. What are two things you notice about this menu?

Answers will vary.

2. What is one thing you wonder?

Answers will vary.

3. What is a suggestion you could give to the owners about their pricing?

Answers will vary.



Highlight student responses that mention the following:
The price difference between adding chips or fries is not consistent and varies from a \$0.45 - \$0.95 difference.
The price to add cheese ranges from \$0.30 - \$0.80.

- Is there a pattern to the differences?
- Is the amount of fries different when the cost is higher?

10.9.2 Guided Instruction: Predicting How Many

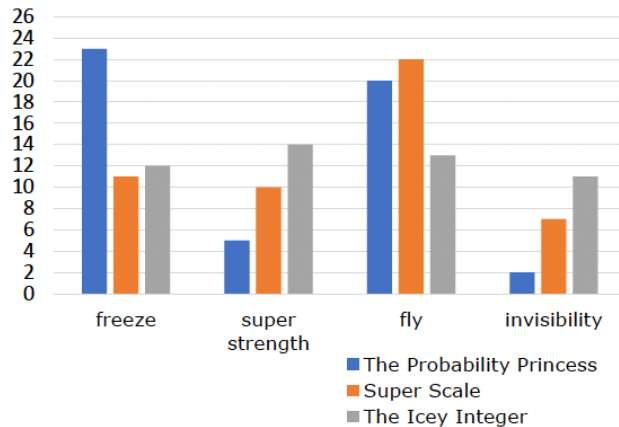
Component Overview: Students learn how to use the data from a random sample to make predictions about a population. They will be introduced to three different strategies to use these data to make predictions. Students will analyze and reflect on each method to determine a strategy that works best for them.



10.9.2 Guided Instruction: Predicting How Many

Results of a random sample can be used to make predictions. The results from a sample of 50 readers of three different comic books are shown.

**Superpowers Comic Readers
Would Like the New Hero to Have**



Suppose the author of *The Probability Princess*, Ashley Grant, is trying to make decisions about the future of the series. She really wants the new hero to have the ability to fly, but she knows there is a larger percentage of readers who want the new hero to have freeze power. As long as the new series can sell at least 3,600 issues in the next five years, Ashley will be able to stay in business.

1. She determines 50 readers of *The Probability Princess* responded to the survey. She then finds the percentage of the 50 readers surveyed who want the new hero to have the ability to fly.

$$\frac{20}{50} = \frac{40}{100} = 0.4 = 40\%$$



The questions in this section are designed to show students how population predictions from survey data can be used to make business decisions.

- Where else might this be a useful calculation?

10.9.2 Guided Instruction: Predicting How Many (continued)

Over the past five years, *The Probability Princess* has had 10,000 loyal readers. Using three different strategies, Ashley uses the data to predict the number of those readers who want to see the new hero have the ability to fly. In each method, she predicts that 4,000 of the 10,000 readers want this.

Equivalent Fraction	Scale Factor	Percent
The fraction of readers who want the new hero to fly is $\frac{20}{50}$. Write equivalent fractions until the denominator shows the population total (in this case 10,000). $\frac{20}{50} = \frac{40}{100} =$ $\frac{400}{1,000} = \frac{4,000}{10,000}$	The fraction of readers who want the new hero to fly is $\frac{20}{50}$. Divide the population by the sample to find the scale factor needed. Then, multiply both the numerator and denominator by this scale factor to write an equivalent fraction. $10,000 \div 50 = 200.$ $\frac{20}{50} \times \frac{200}{200} = \frac{4,000}{10,000}$	The fraction of readers who want the new hero to fly is $\frac{20}{50}$. Find the percent of the sample for the given condition. Then multiply the population by the percent. $20 \div 50 =$ $0.4 \text{ or } 40\%$ $10,000 \times 0.4 = 4,000$ OR $40\% = \frac{40}{100} = \frac{2}{5}$ $10,000 \times \frac{2}{5} = 4,000$



It is important that students understand how each of these strategies work. The discussion questions are meant to provide scaffolding.

10.9.2 Guided Instruction: Predicting How Many (continued)

2. Explain whether one of Ashley's strategies seems to be better than the others for solving this particular problem.

Answers will vary. All three strategies work equally well. If I were wanting to solve this problem mentally, equivalent fractions would be the easiest way for me to do so.

3. What is the benefit of solving a problem using two different strategies?

Answers will vary. When you use multiple strategies to solve your problem you are able to check your work. When you use multiple strategies you can understand the problem in a different way.

4. Next, Ashley finds the percent of the 10,000 issues they need to sell in the next five years. Complete the work shown below.

$$\frac{3,600}{10,000} = \frac{36}{100} = 0.36 = 36\%$$

Ashley estimates that if the new hero has the ability to fly, at least 4,000 readers will continue to read and purchase *The Probability Princess*. She estimates that this will lead to each of those 4,000 readers purchasing at least one comic. She also thinks it's likely that readers who did not want the new hero to have the ability to fly may still buy issues in the next five years. Because 4,000 is greater than the 3,600 issues they need to sell to stay in business, Ashley decides to give the next hero the ability to fly.

5. Explain why you agree or disagree with Ashley's decision.

Answers will vary. I agree with Ashley's decision. She used proportional reasoning – scale factor and equivalent fractions to prove her thinking. She knows that the 4,000 minimum readers is larger than the amount they need to break even.



Students may struggle to answer Question 5. Consider asking the following questions to further scaffold and guide student thinking:

- How many issues must Ashley sell in the next five years to stay in business? (At least 3,600.)
- How many of her loyal readers does she predict will still buy the series if the new hero has the ability to fly? (At least 4,000 of the 10,000 loyal readers will still buy the series if the new hero has the ability to fly.)
- How does she determine if she can keep enough readers to keep in business? (Since 40% of the readers surveyed want the hero to fly, she figures 40% of the loyal readers will continue to purchase the series, which is 4,000 people and should translate into at least 4,000 issues being sold over the next five years. Since 4,000 is greater than 3,600, or the number of copies she needs to sell to stay in business, Ashley concludes she can give the new hero the ability to fly.)

10.9.3 Exploration: Reaction Times

Component Overview: Students explore how to make predictions using probability. Students are given the opportunity to work collaboratively using the strategy of their choice to make their predictions based on sample data.



10.9.3 Exploration: Reaction Times

The track coach at Yulee High School in Nassau County, FL needs a student whose reaction time is less than 0.4 seconds to help out with timers at track meets. All the twelfth graders in the school measured their reaction times.

A random sample of 32 students' reaction times has been recorded in the table.

Reaction Times of Less Than 0.4 Seconds	Reaction Times of 0.4 Seconds or More
22	10

1. What fraction of the students in the random sample have a reaction time less than 0.4 seconds?

$$\frac{22}{32} = \frac{11}{16}$$

2. Explain or show how to estimate the percentage of selected twelfth graders from this school who had a reaction time of less than 0.4 seconds.

A random sample of twelfth graders would have a

$$\frac{22}{32} = \frac{11}{16} = 0.6875 = 68.75\% \text{ chance of a reaction time of less than 0.4 seconds.}$$

3. There are 125 twelfth graders at this school. How many of the twelfth graders are expected to have a reaction time of less than 0.4 seconds?

To find this answer we would use scale factor.

$$125 \times 22/32 = 85.9375 \text{ This can be rounded to 86 students who are likely to have reaction times less than 0.4 seconds.}$$



This Exploration is scaffolded to have students discuss their thinking and strategies as they solve. Students may use different strategies to answer this question.



*Students often struggle with estimation. Consider asking guiding questions to further scaffold student thinking:
How can we determine if an answer is reasonable or not?
What might an unreasonable answer be?
How can you explain differences in answers?*

10.9.3 Exploration: Reaction Times (continued)

4. Suppose another group in your class comes up with a different estimate than yours for the previous question. What is another estimate that would be **reasonable**?

Another reasonable estimate could be 87.5 if they rounded the percentage of twelfth graders with a reaction time of less than 0.4 seconds to the nearest tenth.

5. Explain what kind of an estimate would be considered **unreasonable**.

An estimate of 100 would be unreasonable. In that instance, rounding to the nearest hundred actually significantly changes the percentage of adjustment. An estimate of 100 would be equivalent to about 8 tenths of the number of total people rather than 7 tenths.



Consider using informal observations as students complete this component to inform potential groupings for the next component, 10.9.4 Guided Practice.

10.9.4 Guided Practice: Making Population Predictions

Component Overview: Students synthesize and apply their understanding from 10.9.2 Guided Instruction and the 10.9.3 Exploration to complete problems on their own (or with a partner). They should be able to use the strategy of their choice and discuss when each strategy is most useful.



10.9.4 Guided Practice: Making Population Predictions

1. Taylor attends Murdock Middle School in Port Charlotte, Florida. She wonders how many students at her school would dye their hair blue if their parents would let them. She surveyed a random sample of 10 students at her school, and 2 of them said they would. Kiran didn't think Taylor's estimate was very accurate, so he surveyed a random sample of 100 students, and 17 of them said they would.
 - a. Complete the table below to compare the accuracy of Taylor and Kiran's random samples. Be sure to show your work or explain your thinking.

	Taylor	Kiran
Fraction of students surveyed who would dye their hair blue	$\frac{2}{10} = \frac{1}{5}$	$\frac{17}{100}$
Percentage of students surveyed who would dye their hair blue	$\frac{2}{10} = \frac{20}{100}$ $= 20\%$	$\frac{17}{100} = 17\%$
Predicted number of the 500 students in the school who would dye their hair blue	$\frac{20}{100} = \frac{100}{500} =$ 100 students	$\frac{17}{100} = \frac{85}{500} =$ 85 students

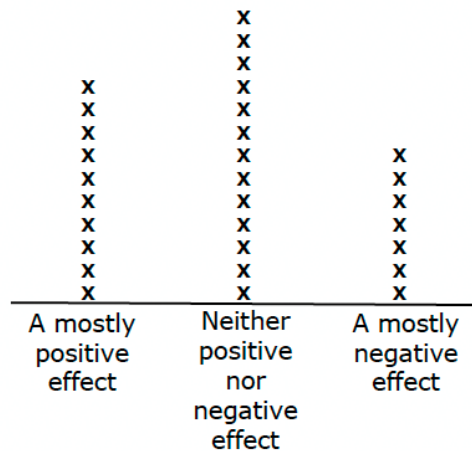
- b. Explain whose random sample likely produced the most accurate estimate.
Answers will vary. Kiran's sample included more students so that is a more accurate estimate.



Either before or after this component, consider having students brainstorm which strategies they most prefer and when they think each strategy is most useful.

10.9.4 Guided Practice: Making Population Predictions (continued)

2. Ms. Jeffrey teaches seventh grade in Miami, Florida. She noticed many of her students talked about social media and wondered about what her students considered to be the impact of all their online activity. She conducted a random survey of 30 seventh-grade students asking them how they describe the effect of social media use on people their own age. The results are shown in the line plot below.



According to a 2018 Pew Research Center of American teenagers, 24% of teen respondents said they believed social media has a mostly negative effect on people their own age.

- a. Explain how this compares to the percentage of Ms. Jeffrey's seventh-grade students who described social media as having a mostly negative effect.
In Ms. Jeffrey's class, there were seven out of thirty students who said they felt social media had a mostly negative effect. $\frac{7}{30} = 0.23 = 23\%$. This is about equal to what the study had found.



For students who finish early, consider prompting them to create an advertisement using this data. Who would need this data? How could they use it to persuade an audience?

10.9.4 Guided Practice: Making Population Predictions (continued)

There were about 1,546,397 teenagers living in the state of Florida in 2018.

- b. Predict the number of teenagers in the state of Florida who considered the effect of social media as mostly positive based on the data Ms. Jeffrey collected.

Answers will vary. The number of students in Ms. Jeffrey's class who considered the report mostly positive were $\frac{10}{30} = \frac{1}{3}$. $\frac{1}{3}$ of 1,546,397 is approximately 515,466 teenagers. Using these numbers, I could also approximate that about $\frac{1}{3}$ of 1.5 million is approximately 500,000 teenagers.

- c. Predict the number of teenagers in the state of Florida who considered the effect of social media as neither positive nor negative based on the data Ms. Jeffrey collected.

The number of students in Ms. Jeffrey's class who considered the report mostly positive were $\frac{13}{30} = 0.433 = 43.3\%$. $0.433 \times 1,546,397 = 669,589.90$. I would predict about 669,590 teenagers responded this way.

- d. Explain at least two factors that could account for differences in these estimates from the actual number of students who would have described the effect of social media in these ways.

Answers will vary. See below.

Why are they different?

- I chose not to round 43.3%, but someone may have rounded it to 40% or 45%.*
- I rounded to the number of one and a half million before dividing by 3. Other people may have wanted to directly find the answer instead of estimating.*



Students may struggle to extend the data to predict the numbers for the entire state of Florida.

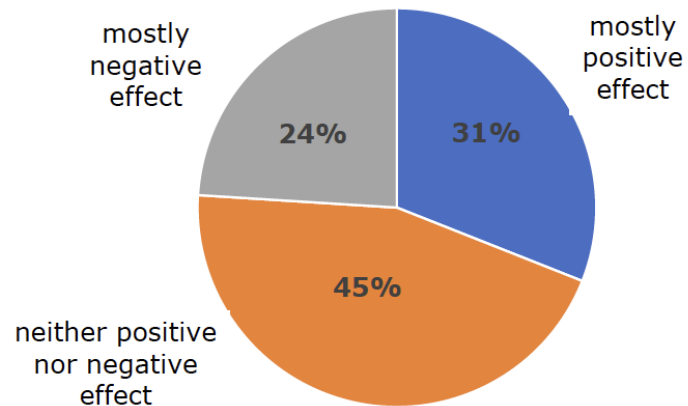
This question provides opportunities for students to consider the ways rounding can impact estimates.

In problems that require several steps, students should not round until the final step. Rounding midstep could significantly change the answer. If not given a place value for rounding, then determine what place value you would like your students to round to. Many assessments have students round to three decimal values. Incorporating the Desmos scientific calculator can add a powerful tool for students who struggle with completing steps on a handheld calculator. The Desmos calculator allows students to input complicated arithmetic that would require several steps on a handheld calculator. This eliminates common, careless errors that many students make.

10.9.4 Guided Practice: Making Population Predictions (continued)

The results of the 2018 Pew Research Study are shown in the circle graph.

Teen Views on the Effect of Social Media on People Their Age



Students are given a visual representation of the data, which lends itself not only to understanding the data, but also to understanding the probability of each event.

- e. The percentage of teens who describe social media as having a positive effect on people their age is ☐ less than ☐ equal to ☐ greater than the percent of students who responded the same in Ms. Jeffrey's survey.

Write an explanation of why this may have happened.

Answers will vary. In Ms. Jeffrey's class, $\frac{1}{3}$ of the students rated social media as positive. In the real study it was 31%. If we had used the more exact measure of 31% we would have gotten a more accurate answer.

10.9.5 Wrap-Up: Growth Mindset

Component Overview: Students reflect on their understanding of the lesson's learning target and consider ways to overcome any barriers in future lessons.



10.9.5 Wrap-Up: Growth Mindset

Complete the following sentences based on today's learning target:

I can use proportional relationships to make predictions about a population.

1. A barrier to my learning today was ...

Answers will vary. I struggled to remember my multiples and factors so finding the equivalent fractions and scale factors was challenging. I struggled with knowing how much to round when estimating.

2. I will try to overcome this by ...

Answers will vary. I can practice multiplication tables and factor trees. I can practice reasonable estimates by using a number line to assess how far my guess is from the original number.

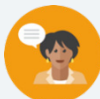


Consider prompting students to use Mark-Up Text to connect their responses in questions 1 and 2 to their learning throughout the lesson. Students can draw a symbol or emoji to represent “barriers,” and mark questions with that symbol, providing even more information for their progress from the lesson.

10.10 Making Population Predictions – Part 2

Lesson Introduction

 Benchmark Focus	MA.7.DP.1.3 Given categorical data from a random sample, use proportional relationships to make predictions about a population.		 Learning Target	I can use proportional relationships to make predictions about a population.
 Benchmark Coherence	Building On: N/A		Working Toward: MA.912.DP.1.4 Estimate a population total, mean, or percentage using data from a sample survey and develop a margin of error through the use of simulation.	
 Essential Questions	- How can data from a sample be used to make predictions about a population? - How can fractions and percentages be used to make population predictions?			
 Lesson Narrative	Students continue their exploration of using proportional relationships to make predictions about populations from sample data. Students complete a formative assessment as Bell Work, then work through analyzing strategies and interpreting solutions under the teacher's guidance. Students then synthesize all they've learned to complete a Collaboration where they discuss and choose the best strategies to determine how to stock a school vending machine based on sample data. After independent practice, students complete a formative assessment as a check for understanding of the learning target.			
 Component Summary	10.10.1 Warm-Up (~5 min.)	Predict the number of fish that will be caught based on previous data (<i>SE p. 158</i>)	10.10.3 Collaboration (10-15 min.)	Collaborate to make predictions about how to stock a vending machine based on student survey data (<i>SE p. 160</i>)
	10.10.2 Guided Practice (10-15 min.)	Discuss and analyze strategies used during Guided Practice (<i>SE p. 158</i>)	10.10.4 Your Turn! (10 min.)	Make predictions about a population based on sample data (<i>SE p. 161</i>)
	10.10.5 Wrap-Up (~5 min.)	Formative assessment of the lesson's learning target (<i>SE p. 162</i>)		
 Resources	Glossary		Materials	
	<u>Key Terms:</u> - Scale factor - Equivalent fraction - Percent <u>Additional Terms:</u> N/A		N/A	
			Additional Preparation	
			N/A	

 Teacher Tips	Common Misconceptions	Things to Consider
	Students may have trouble choosing the strategy that is best for each situation or struggle to reason about which is the most efficient.	- At this time, the focus in problem-solving for these types of questions is NOT on using proportions. Unit 12 will focus on setting up and solving proportions. The focus in this unit is on using strategies such as equivalent fractions, scale factor, and percentages. - In problems that require several steps, students should not round until the final step. Rounding mid-step could significantly change the answer. If not given a place value for rounding, then determine what place value you would like your students to round to. Many assessments have students round to three decimal values. Incorporating the Desmos scientific calculator can add a powerful tool for students who struggle with completing steps on a handheld calculator. The Desmos calculator allows students to input complicated arithmetic that would require several steps on a handheld calculator. This eliminates common careless errors that many students make.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Think-Pair-Share 10.10.2 Guided Practice positions students to consider and reason about which strategy makes the most sense for them to use and why. Encourage students to build on their partner’s responses or make changes to theirs if they find that one strategy may be more beneficial than their original.	MA.K12.MTR.4.1: Discussion and Reflection 10.10.3 Collaboration provides students an opportunity to work together and present a variety of ideas for how to use the sample data to solve a problem. Consider providing opportunities for students to present their strategies and ideas to the class so that they can have the opportunity to analyze the mathematical thinking of others. For example, class time could be spent here and the 10.10.4 Your Turn! could be assigned for homework instead.
 Differentiate Instruction	Some students may be overwhelmed by choosing from the three strategies, or it may be the case that only certain strategies make sense to them. Students can be given two choices instead of three. For those students who are overwhelmed by having to choose from even two strategies, allow them to use one strategy. Students will continue working with proportional reasoning and these strategies throughout the upcoming units, so it is not necessary that they understand all strategies well at this point in instruction.	
	Additional differentiation opportunities are denoted throughout the lesson guide.	
Lesson Notes:		

10.10.1 Warm-Up: Bell Work

Component Overview: Students practice skills from this unit by scaling down a population of fish.



10.10.1 Warm-Up: Bell Work

Antonia and her aunt go fishing at Lake Okeechobee every Sunday. For the past four Sundays, Antonia has kept track of what kinds of fish they caught.

Type of Fish	Bluegill	Bream	Crappie	Largemouth Bass
Number of Fish Caught Last Month	10	12	20	8
Expected Number of Fish to Be Caught Next Sunday	3	3	5	2



Consider positioning students to discuss their strategies for determining the number of fish to be caught on just one Sunday. This gives students an opening to what they'll be working on in this lesson and connects to the idea of reasonable versus unreasonable predictions from the prior lesson.

1. Complete the table to show what type and how many fish Antonia and her aunt are likely to catch next Sunday when they go fishing.

10.10.2 Guided Practice: Making Population Predictions

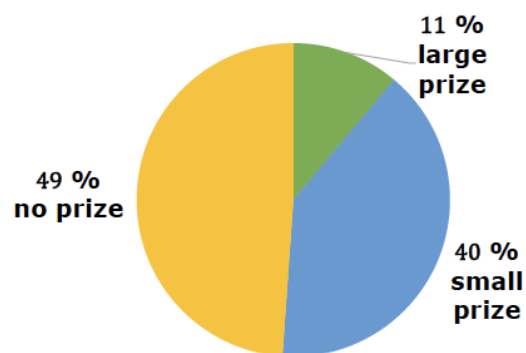
Component Overview: Students continue to build upon their knowledge of using sample data to make predictions about populations. They begin by analyzing a student's work and strategies before applying the strategy of their choice to a different problem.



10.10.2 Guided Practice: Making Population Predictions

1. Ella gathered data from a Rubber Duck Pool Game. The circle graph below shows the percentage of people who won a large prize, a small prize, and no prize.

Prizes Won in Rubber Duck Game



There are 180 rubber ducks floating in the pool. The person playing the game takes out one duck and looks for a mark on the bottom of it.

- If there is a blue mark, the person wins a small prize.
- If there is a green mark, the person wins a large prize.
- If there is no mark, the person does not win a prize.

Ella uses her data to predict how many of each kind of duck are in the pool using three different strategies. Her work is shown.



Consider prompting students to mark the text using the Mark-Up strategy as they read and understand the context.

10.10.2 Guided Practice: Making Population Predictions (continued)

Method 1	Method 2	Method 3
40% of all ducks	11% of all ducks	49% of all ducks
$\frac{40}{100} = \frac{4}{10} = \frac{72}{180}$	$0.11 \times 180 = 19.8$	$\frac{49}{100} \times \frac{0.56}{0.56} = \frac{27.44}{180}$
So, 72 ducks have a blue mark.	So, 20 ducks have a green mark.	So, 27 ducks have no mark.
<i>Equivalent fraction</i>	<i>Percent</i>	<i>Scale factor</i>

- a. Ella's solution ☐ is ☐ is not correct because

- ☐ the number of ducks with green marks is less than the number of ducks with blue or no marks.
- ☐ the number of ducks with no marks is less than the number of ducks with blue marks.
- ☐ Ella's solution shows a total of 180 ducks.

- b. Write an explanation for why you agree or disagree with Ella's solution. Use specific examples from Ella's strategies to justify your opinion.

I do not agree with Ella's solution. Her answer does not total to 180. If you add the terms together, you have 119 ducks. It looks like her mistake was in the third method. She should have multiplied the numerator and denominator by a scale factor of 1.8, not 0.56. This would have resulted in 88.2, or 88 ducks.

- c. Compare your explanation with your partner's. Discuss any differences and make changes if necessary. *Answers will vary.*
- d. Ella used three different solution strategies. In the bottom of each column in Ella's table, label each strategy with the correct name from the descriptions.



Students may struggle to determine whether or not Ella's solution is correct. Encourage students to consider the following questions to reason about Ella's response and continue to use the pie chart as a reference:

- What do Ella's answers add up to? What should the total be?
- Even before completing the calculations, what group should the largest number of ducks be? How do you know? The smallest amount? Do these match Ella's answers?

10.10.2 Guided Practice: Making Population Predictions (continued)



When using a **scale factor**, first divide the population by the sample to find the scale factor, then multiply the numerator and denominator of the sample fraction by the scale factor.



When using an **equivalent fraction**, write multiple fractions equal to the sample fraction until the denominator shows the total population.



When using a **percent**, multiply the total population by the decimal or fraction form of the percent of the sample.

2. Bianca picks blocks out of a bag one at a time. She picks out a block, records the color, and puts the block back in the bag. She cannot see the blocks, but she knows there are 6 blocks in the bag. Of the 50 times she picked a block out of the bag, she picked a blue block 29 times, a red block 10 times, and a yellow block 11 times.

- a. Write ratios to show the number of times Bianca picked a blue, red, and yellow block out of the bag.

Blue $\frac{29}{50}$ Red $\frac{10}{50} = \frac{1}{5}$ Yellow $\frac{11}{50}$

- b. Explain which strategy will work best to predict how many of the 6 blocks are blue, red, and yellow.

Answers will vary.

Sample answer: The strategy of percent would be the best strategy to use in this case since we already have the fraction form of the percent from the sample from part a.



Consider having students label each of Ella's strategies with one of these names. Labeling each of Ella's strategies provides a reference point for students to return to as they work to solve similar problems on their own.



Think-Pair-Share

Position students to discuss which strategy they think will work best and why before completing question 2b in writing. This will allow students to reason about each of the strategies and potentially hear why another strategy might be more efficient or make more sense. Students have the opportunity to solidify their thinking before putting it in writing.

10.10.2 Guided Practice: Making Population Predictions (continued)

- c. Using the strategy you selected, how many of the 6 blocks are blue, red, and yellow? Show your work.

Answers will vary.

Percent Strategy:

Blue: $\frac{29}{50} \times 6 = 3.48 \approx 4$

Red: $\frac{10}{50} \times 6 = 1.2 \approx 1$

Yellow: $\frac{11}{50} \times 6 = 1.32 \approx 1$

Therefore, there is 1 red block, 1 yellow block and 4 blue blocks.

- d. Explain whether the strategy you chose was the best strategy to use, or if another would have worked better.

Answers will vary. The strategy that I chose was the best because we already knew the fraction form of the percent from the sample in part a. Using equivalent fractions could be tricky because the least common multiple for 6 and 50 is 300. Using scale factor would require additional work to first determine that the scale factor is $\frac{6}{50} = 0.12$, and then multiply the scale factor by numerator and denominator of the sample fraction.

10.10.3 Collaboration: Stocking a Vending Machine

Component Overview: Students work collaboratively to use survey data to make predictions in regard to stocking a school vending machine. Students are given the opportunity to choose their own strategy when solving and to discuss with their partners their mathematical processes.



10.10.3 Collaboration: Stocking a Vending Machine

Students at Sanford Middle School are upset the school vending machine is always sold out of their favorite items. The student council decides to survey students to find a better way to stock the vending machine. They randomly sample 60 students. Each student surveyed picked their 5 favorite items from the vending machine. The results are shown in the table.

Item	Number of Times Chosen in Survey	Number that Should be Stocked
Reese's	30	45
Skittles	24	30
Starburst	21	15
Snickers	42	75
Pretzels	24	30
Chips	30	45
Takis	39	75
Gum	36	60
M&Ms	30	45
Trail Mix	24	30

The vending machine at school has 30 different snack compartments, each with 15 metal spirals for each snack. The vending machine can hold 450 total snacks.

1. Work with your partner to decide how many of each snack should be stocked in the vending machine. Complete the table with your decision.

Your work will be assessed using these criteria.

- Each of the strategies: scale factor, equivalent fraction, and percent are used. If one of the strategies is not used, there is an explanation for why the strategy was not used.
- It is clear, either from your work or a written explanation, why you have stocked the vending machine the way you did.



Students may have trouble choosing the strategy that is best for each situation. Since all of the strategies work in all situations, they have flexibility in their choice.



Consider the different ways students might approach this problem and circulate as they work to identify unique strategies. There is not one correct answer to this problem.

10.10.3 Collaboration: Stocking a Vending Machine (continued)

- The number of each snack stocked in the vending machine is a direct reflection of the survey results.

Item	Number of Times Chosen in Survey	% of Students Who Selected Item in Top Five Favorite	# of Rows Each Item Should Have	Number that Should be Stocked
Reese's	30	$\frac{30}{60} = \frac{1}{2} = 50\%$	3	$3 \times 15 = 45$
Skittles	24	$\frac{24}{60} = \frac{2}{5} = 40\%$	2	$2 \times 15 = 30$
Starburst	21	$\frac{21}{60} = \frac{7}{20} = 35\%$	1	$1 \times 15 = 15$
Snickers	42	$\frac{42}{60} = \frac{7}{10} = 70\%$	5	$5 \times 15 = 75$
Pretzels	24	$\frac{24}{60} = \frac{2}{5} = 40\%$	2	$2 \times 15 = 30$
Chips	30	$\frac{30}{60} = \frac{1}{2} = 50\%$	3	$3 \times 15 = 45$
Takis	39	$\frac{39}{60} = \frac{13}{20} = 65\%$	5	$5 \times 15 = 75$
Gum	36	$\frac{36}{60} = \frac{3}{5} = 60\%$	4	$4 \times 15 = 60$
M&Ms	30	$\frac{30}{60} = \frac{1}{2} = 50\%$	3	$3 \times 15 = 45$
Trail Mix	24	$\frac{24}{60} = \frac{2}{5} = 40\%$	2	$2 \times 15 = 30$

Use the survey results to find the percentage of surveyed students who selected the snack in their top five favorite. Based on preference, determine how many rows each item should have in the vending machine. In the table above, the rows were determined by the following:

$0\% - 35\% = 1 \text{ row}$
 $36\% - 45\% = 2 \text{ rows}$
 $46\% - 55\% = 3 \text{ rows}$
 $56\% - 65\% = 4 \text{ rows}$
 $66\% - 100\% = 5 \text{ rows}$

Since we know there are 15 snacks in each row of the vending machine, use the scale factor of 15 to determine to total number of each item.



Further scaffold this section for students as they work if needed. Consider posting these questions somewhere for students to reference or circulating and positioning students to consider and discuss as you see fit:

- What do you notice about the size of the sample compared to the size of the population?
- How many times larger/smaller is the population than the sample?
- What number can be multiplied by the sample size to get the total in the population?
- If the gum was chosen 42 out of 300 times, what percentage is that? (This means gum should make up 14% of the items in the vending machine.)
- How would we determine how many spots out of 450 the gum would take up? ($\frac{42}{300} = \frac{21}{150} = \frac{63}{450}$ of the items in the vending machine.)

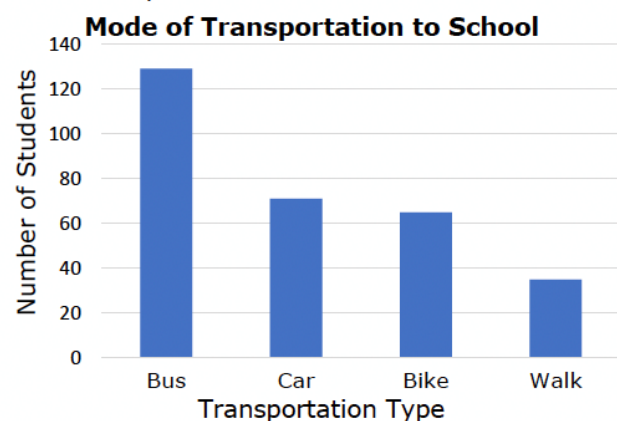
10.10.4 Your Turn!

Component Overview: Students work independently to make predictions about a population using sample data related to transportation to school.



10.10.4 Your Turn!

1. Suwannee County School District is studying changes that will need to be made to the transportation department when building new schools. To determine this, it is important to understand how students get to school. The school district has a population of 5,959 students, so a random sample of approximately 5% of the students in each school was conducted. The results of the survey are shown in the bar chart.



- a. Show how to predict the number of students in the district who take the bus to get to school.
To predict the number of students who take the bus to school, round the total number of students (5,959) to a friendlier number (6,000) and then multiply by 0.05 or 5% which results in 300 students who were surveyed.
From the graph we know that about 130 of the approximately 300 students surveyed take the bus to school.
The scale factor of total number of students in the district to the students sampled is $\frac{6,000}{300} = 20$. So, we can predict that roughly $20 \times 130 = 2,600$ students in the district ride the bus to school.



Students may struggle to choose a strategy to predict the number of students who take the bus or ride their bikes to school. Remind students of the three strategies they have been practicing throughout this lesson and encourage them to use the strategy that makes the most sense to them.

10.10.4 Your Turn! (continued)

- b. Show how to predict the number of students in the district who ride their bike to school.

To predict the number of students who ride their bike to school, we can use the information from part a. that roughly 300 students were surveyed, and the scale factor of students surveyed to the total number of students in the district is 20.

From the graph, we know $\frac{65}{300}$ students surveyed ride their bike to school. So, we can predict that roughly $20 \times 65 = 1,300$ students in the district ride their bike to school.

- c. Whether or not a student can take the bus to school depends on how close they live to the school. Explain how this data set does not take this into consideration and what effect that might have on the district's decision.

Answers will vary. This data set does not include information about how far the students are traveling to get to school. There may be students who ride their bike who are actually within the zone for bus riding but prefer to get the exercise of riding their bikes. It may be useful for the district to conduct a follow up amongst students who are far enough away to be bussed to school to better assess their population size.



If students are struggling to answer question 1c, consider posing the following questions:

- *Does this data set show how far students travel to get to school?*
- *Why might students ride their bikes? Is it possible that students who ride their bikes live in the area where they could take the bus?*
- *Why might the district want to know how far the students are traveling from?*
- *Could the location of a new school change who lives close enough to bike/walk versus who may need to take the bus?*

10.10.5 Wrap-Up: Ticket Out the Door

Component Overview: Students will complete a Ticket Out the Door demonstrating their understanding of making predictions using probabilities from a data sample. This activity will provide instructional data and therefore should be completed independently.



10.10.5 Wrap-Up: Ticket Out the Door

Lin has one month before the seventh-grade class presidential election, and she wants to determine how likely she is to win. Since she can't ask all 665 seventh graders who they are voting for, she conducts a random sample of 105 seventh-grade students. The results of the survey are in the table.

Candidate	Number of People Planning to Vote for the Candidate
Lin	42
Beslinda (Lin's opponent)	30
Undecided	33

1. Use either percent, scale factor, or equivalent fraction to predict how many of the 665 total seventh-grade students currently plan on voting for Lin. Show your work.

Scale Factor	Percent	Equivalent Fraction
$\frac{665}{105} = 6\frac{1}{3}$ $6\frac{1}{3} \times 42 = 266$	$\frac{42}{105} = 40\%$ $0.4 \times 665 = 266$	$\frac{42}{105} = \frac{2}{5} \times \left(\frac{133}{133}\right)$ $= \frac{266}{665}$

2. Use another strategy to predict how many of the total seventh-grade voters are still undecided.

Scale Factor	Percent	Equivalent Fraction
$\frac{665}{105} = 6\frac{1}{3}$ $6\frac{1}{3} \times 33 = 209$	$\frac{33}{105} = 31.4286\%$ $0.314286 \times 665 = 209$	$\frac{33}{105} = \frac{11}{35} \times \left(\frac{19}{19}\right)$ $= \frac{209}{665}$



MA.K12.MTR.2.1: Multiple Representations
Encourage students to use creative methods to represent their reasoning using models or diagrams. For example, students could use a double number line or tape diagram to represent equivalent fractions.

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